



US012417363B2

(12) **United States Patent**
Pearson et al.

(10) **Patent No.:** **US 12,417,363 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **TERMINALS AND METHODS OF USE AND MANUFACTURE THEREOF**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Metropolis IP Holdings, LLC**, Santa Monica, CA (US)

7,075,427 B1 * 7/2006 Pace B61L 23/06
246/126

(72) Inventors: **Gregory Scott Pearson**, Nashville, TN (US); **Gregory Richard Mason**, Iowa City, IA (US)

8,845,125 B1 * 9/2014 Lumsden B61L 29/04
362/249.02

(73) Assignee: **Metropolis IP Holdings, LLC**, Santa Monica, CA (US)

2004/0226993 A1 11/2004 Fulcher et al.
2011/0145141 A1 * 6/2011 Blain B60L 53/65
235/382

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

2011/0203901 A1 8/2011 King et al.
2012/0283012 A1 * 11/2012 Bainbridge G07F 17/3216
463/30

2012/0292385 A1 11/2012 Mackay et al.
2013/0117078 A1 5/2013 Weik, III et al.
2023/0237884 A1 * 7/2023 Qadri G06Q 20/18
235/383

(21) Appl. No.: **18/119,760**

FOREIGN PATENT DOCUMENTS

(22) Filed: **Mar. 9, 2023**

KR 10-1537753 B1 7/2015
WO WO-2024030701 A1 * 2/2024

(65) **Prior Publication Data**

US 2023/0289542 A1 Sep. 14, 2023

OTHER PUBLICATIONS

International Search Report and Written Opinion issued in PCT/US2023/014923 dated Jun. 29, 2023, 9 pgs.

(Continued)

Related U.S. Application Data

(60) Provisional application No. 63/318,014, filed on Mar. 9, 2022.

(51) **Int. Cl.**

G06K 7/10 (2006.01)
G06K 7/14 (2006.01)
H05K 5/00 (2006.01)
H05K 5/02 (2006.01)

Primary Examiner — Toan C Ly

(74) *Attorney, Agent, or Firm* — Fenwick & West LLP

(52) **U.S. Cl.**

CPC **G06K 7/10366** (2013.01); **G06K 7/1413** (2013.01); **H05K 5/0018** (2022.08); **H05K 5/0213** (2013.01)

(57)

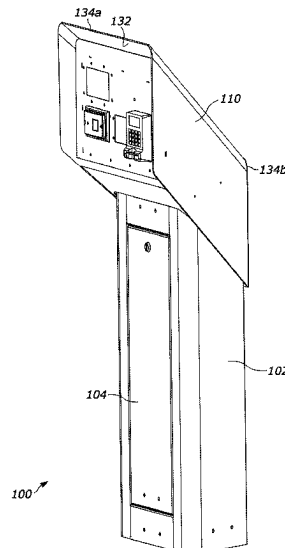
ABSTRACT

Devices and methods are provided for use in controlling access to a parking garage or other restricted area. More specifically, the device relates to a parking terminal to control access to vehicles in a parking area and the method relates to installing and servicing the parking terminal in a desired location without the need for specialized tools.

(58) **Field of Classification Search**

CPC G06K 7/10366
USPC 235/440, 451
See application file for complete search history.

19 Claims, 36 Drawing Sheets



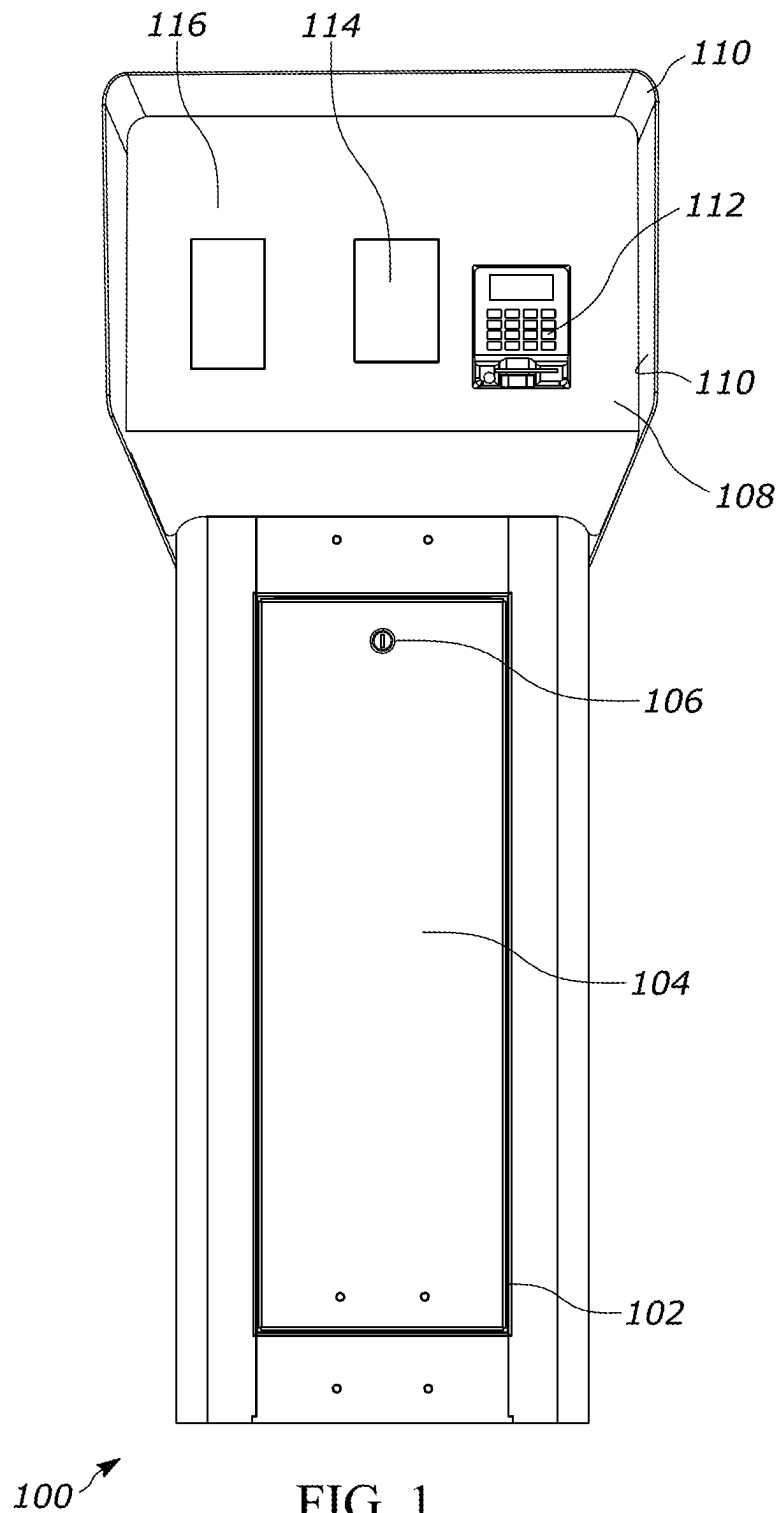
(56)

References Cited

OTHER PUBLICATIONS

International Preliminary Report on Patentability and Written Opinion of the International Searching Authority dated Sep. 10, 2024; 6 pgs.

* cited by examiner



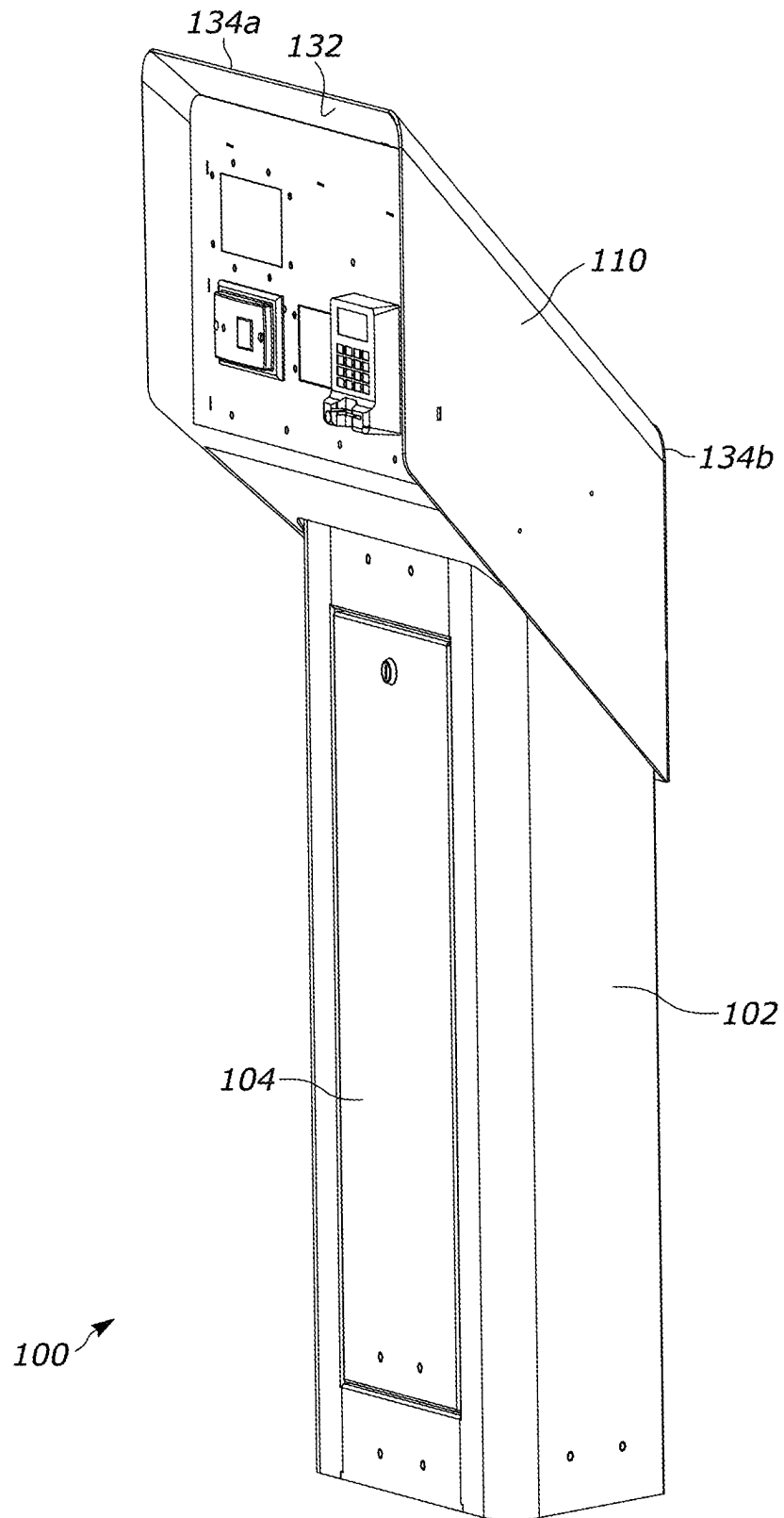


FIG. 2

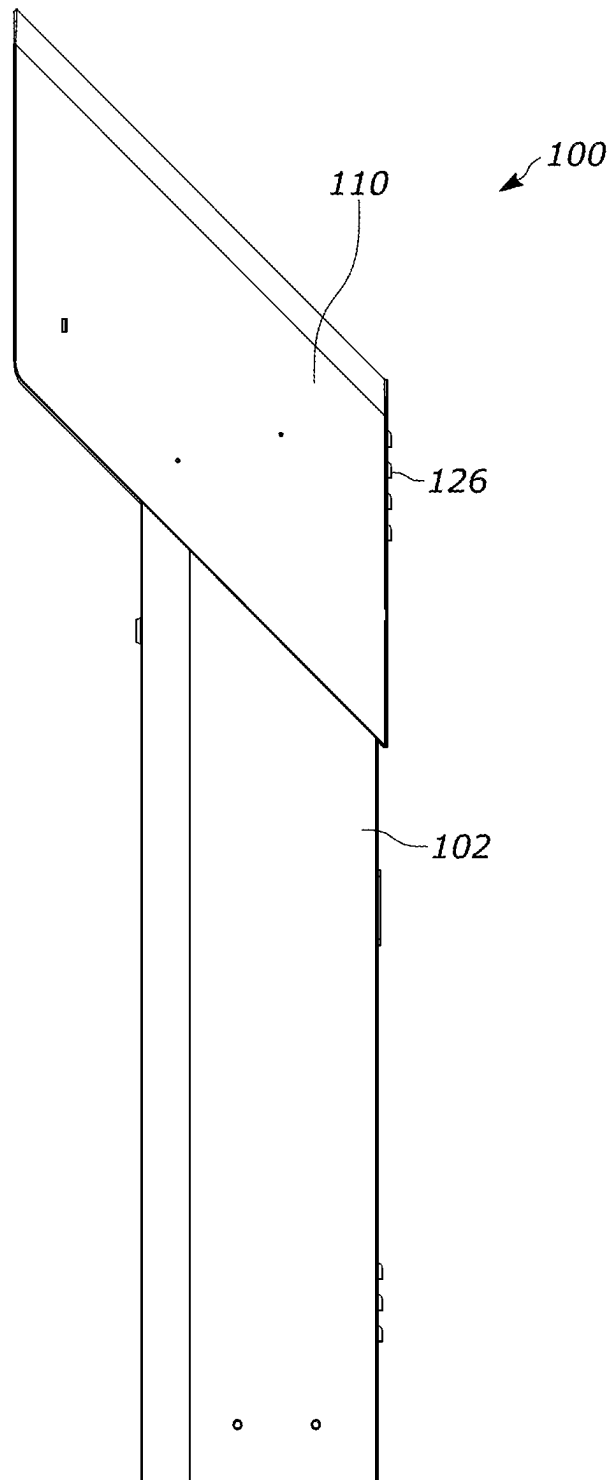


FIG. 3

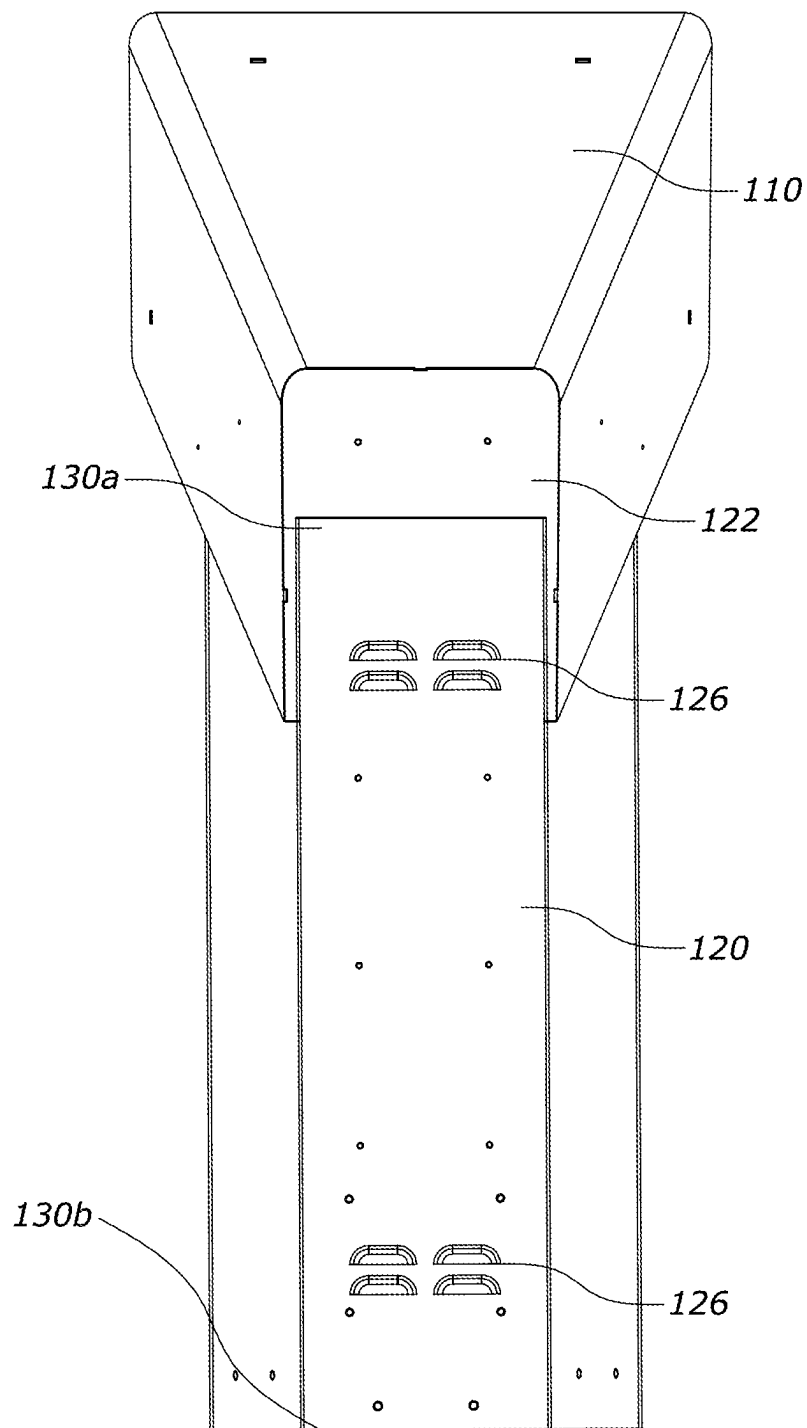


FIG. 4

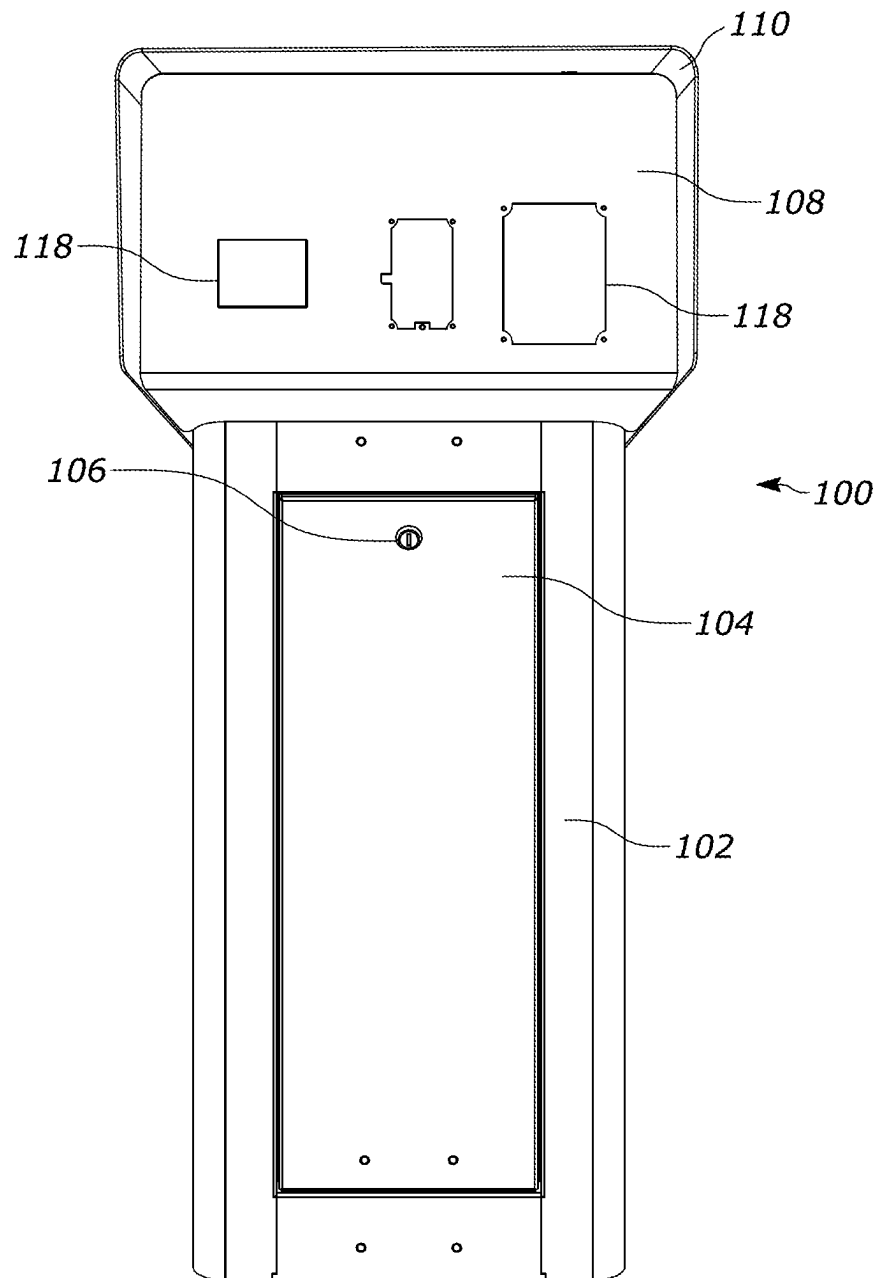


FIG. 5A

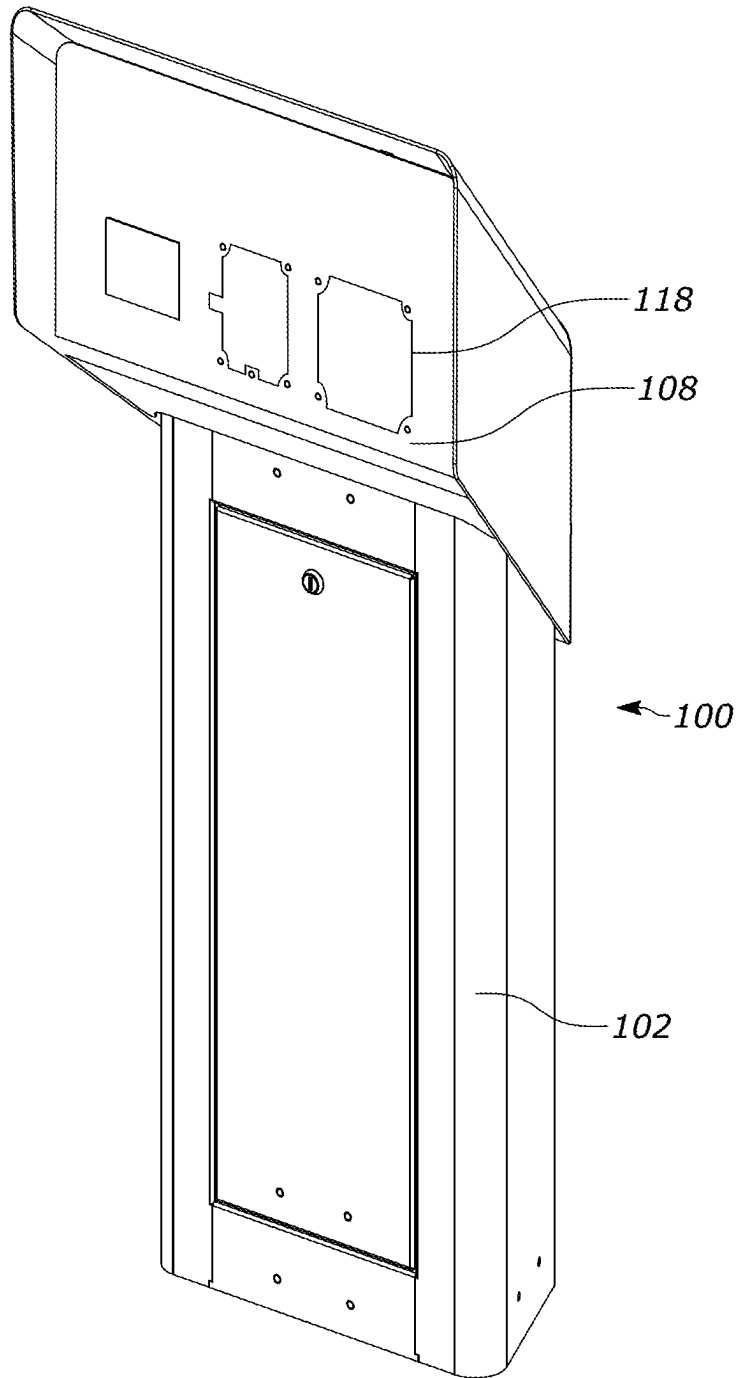


FIG. 5B

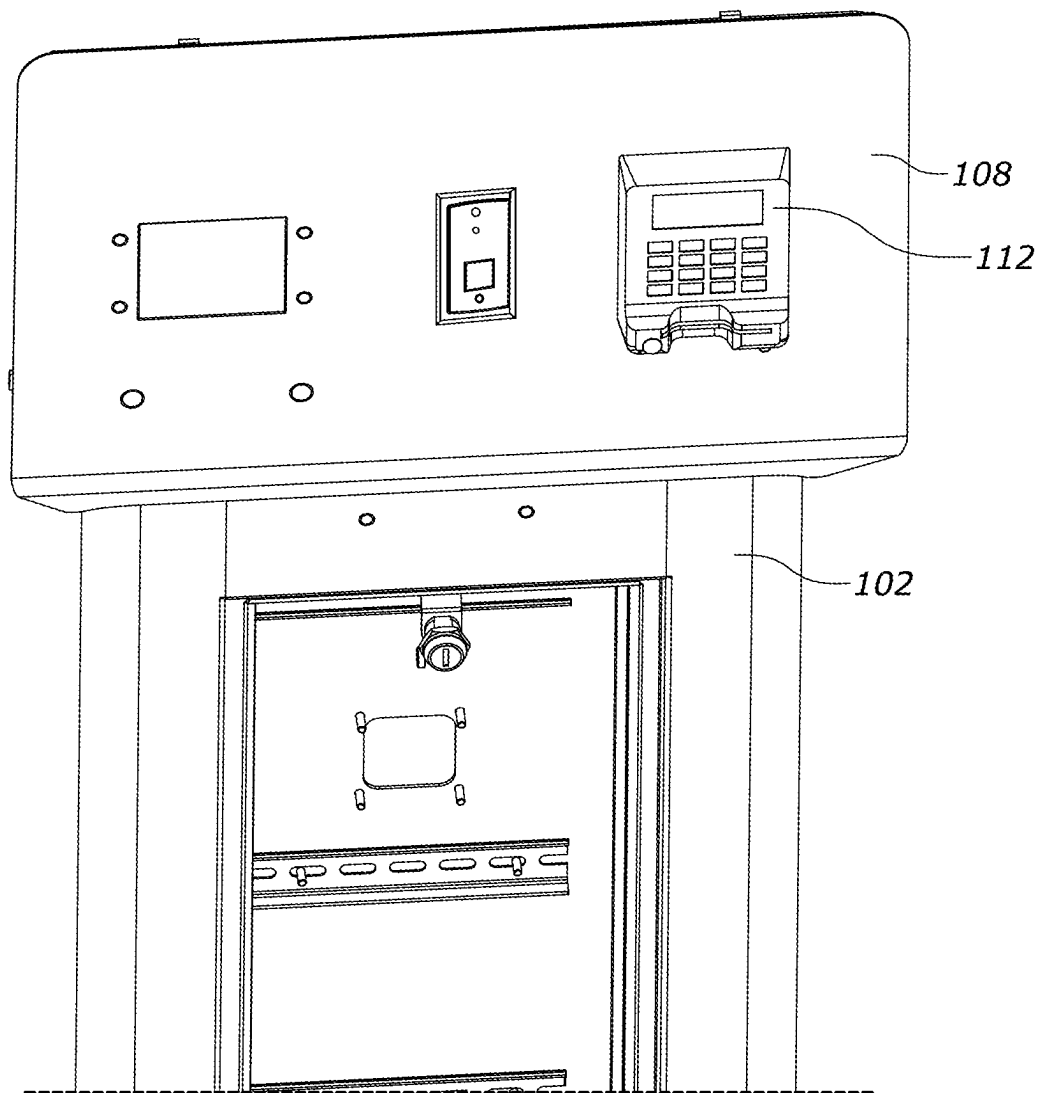


FIG. 6

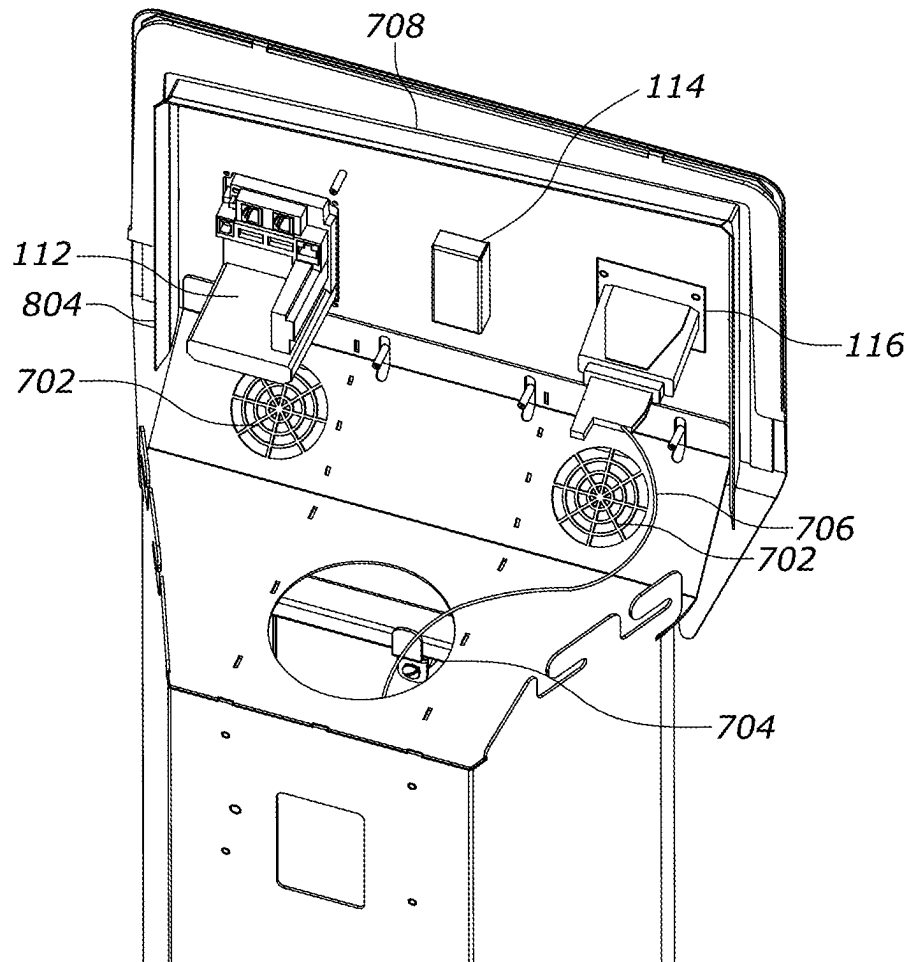


FIG. 7

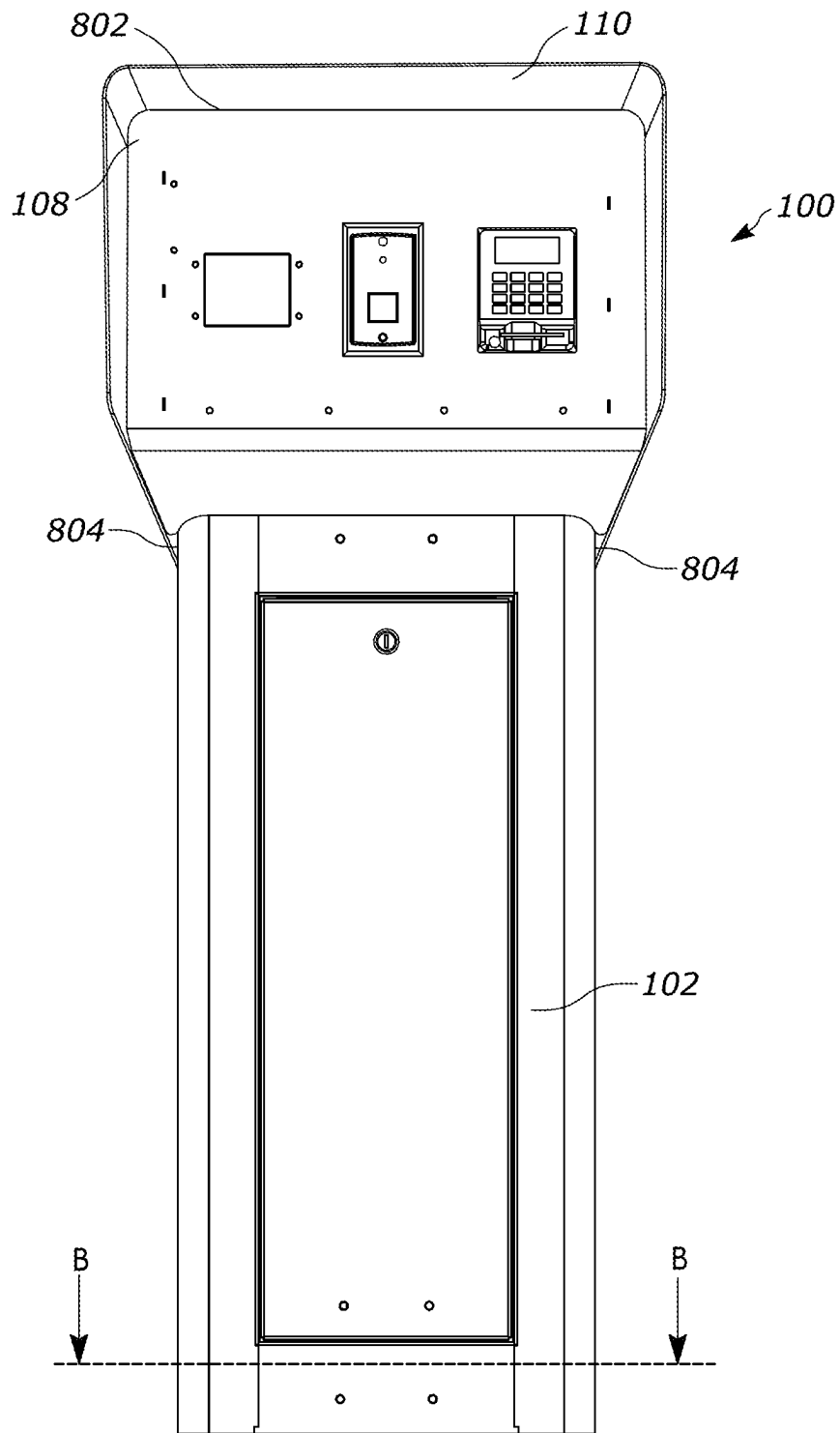


FIG. 8

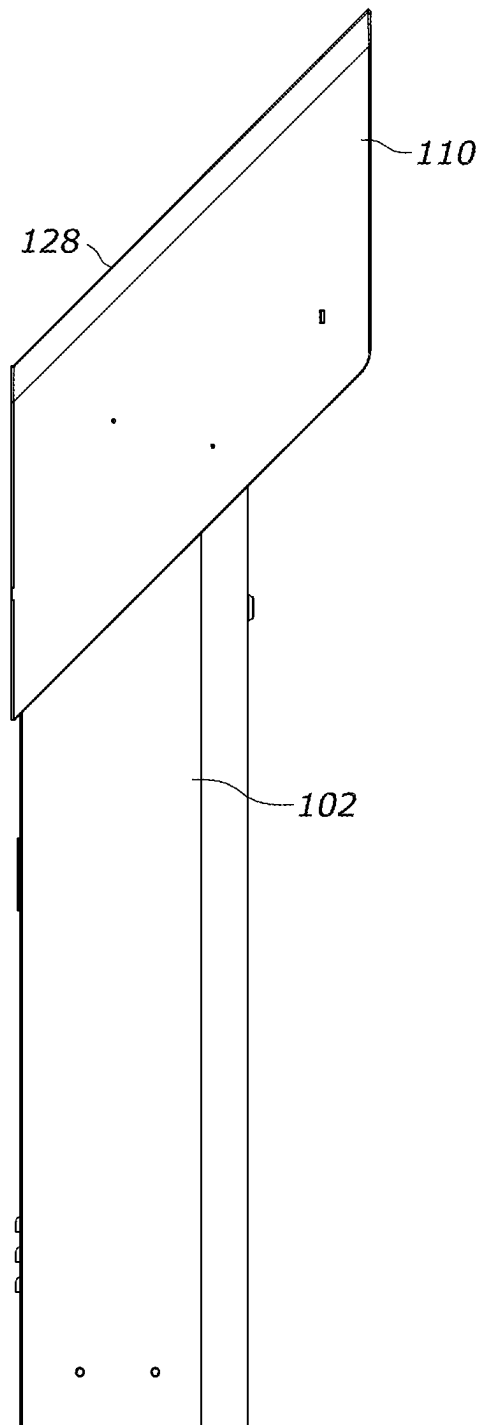


FIG. 9

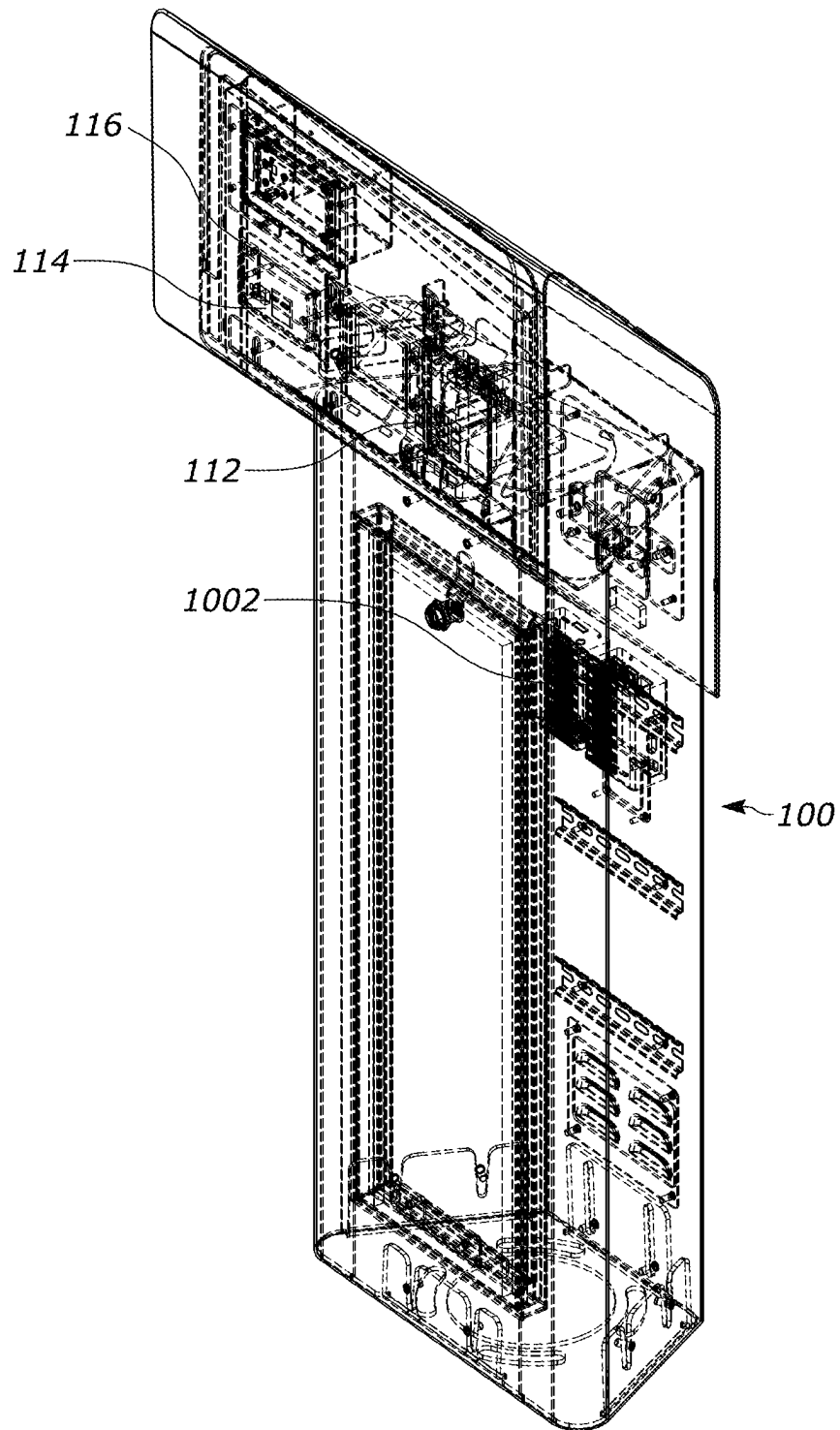


FIG. 10

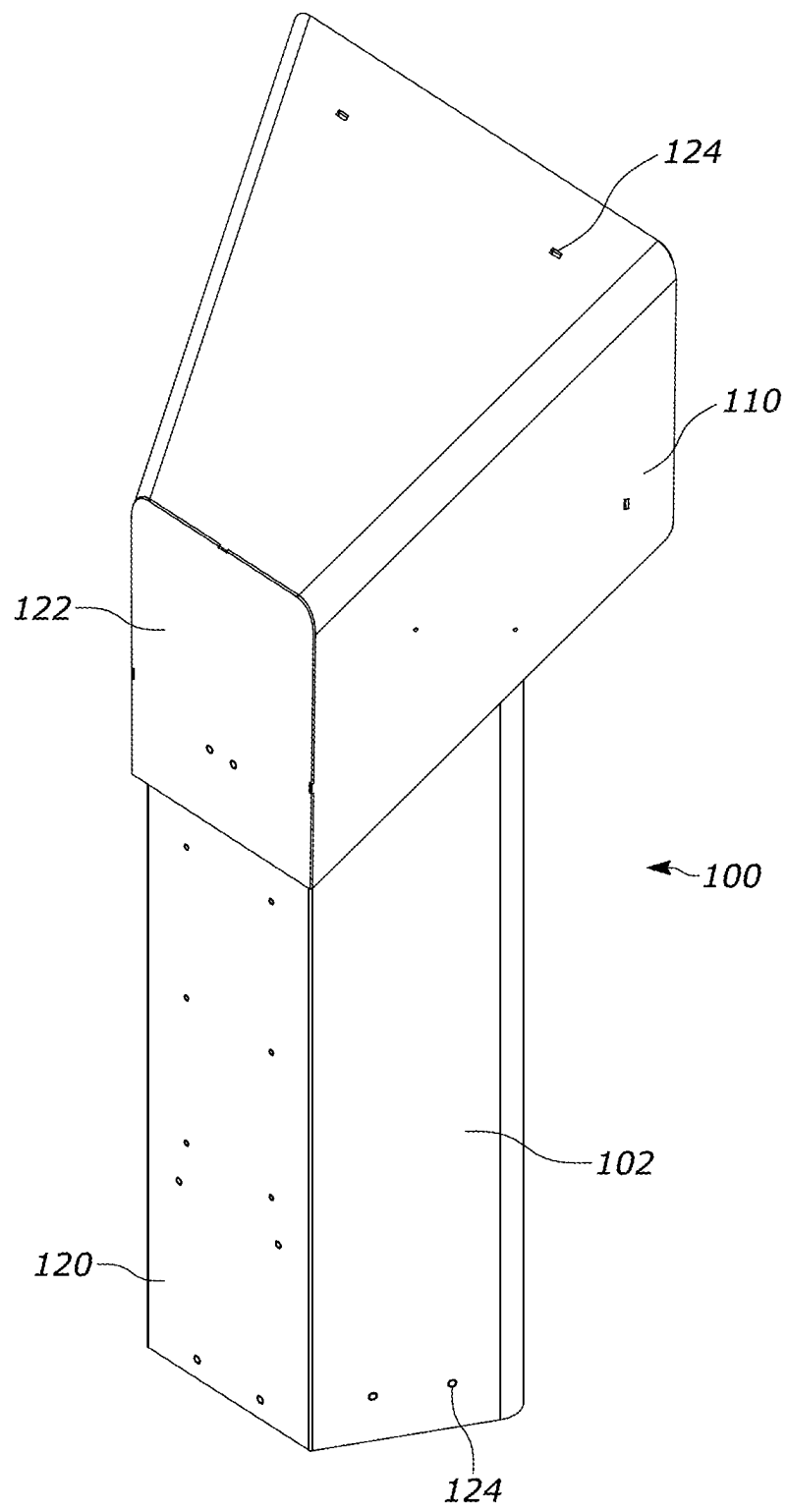


FIG. 11

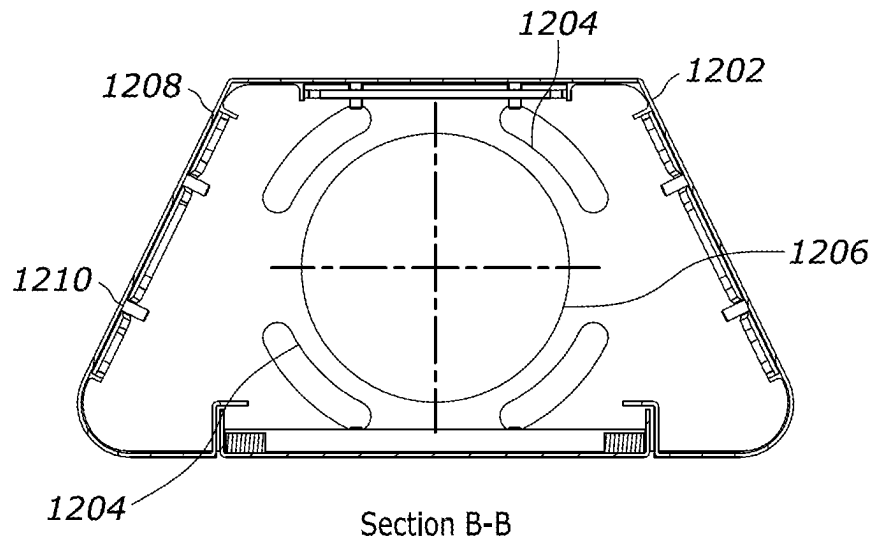


FIG. 12

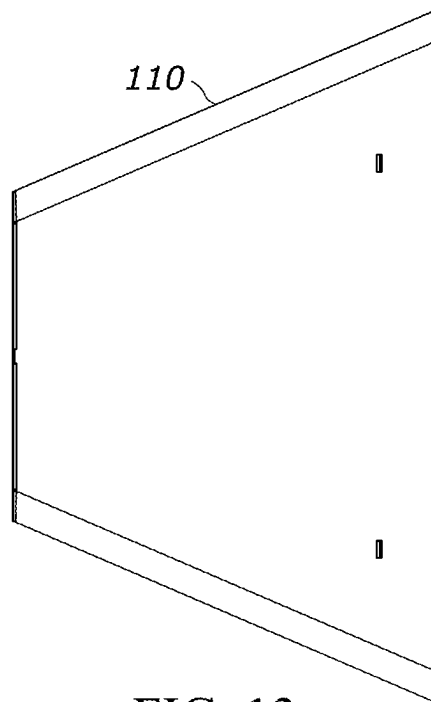


FIG. 13

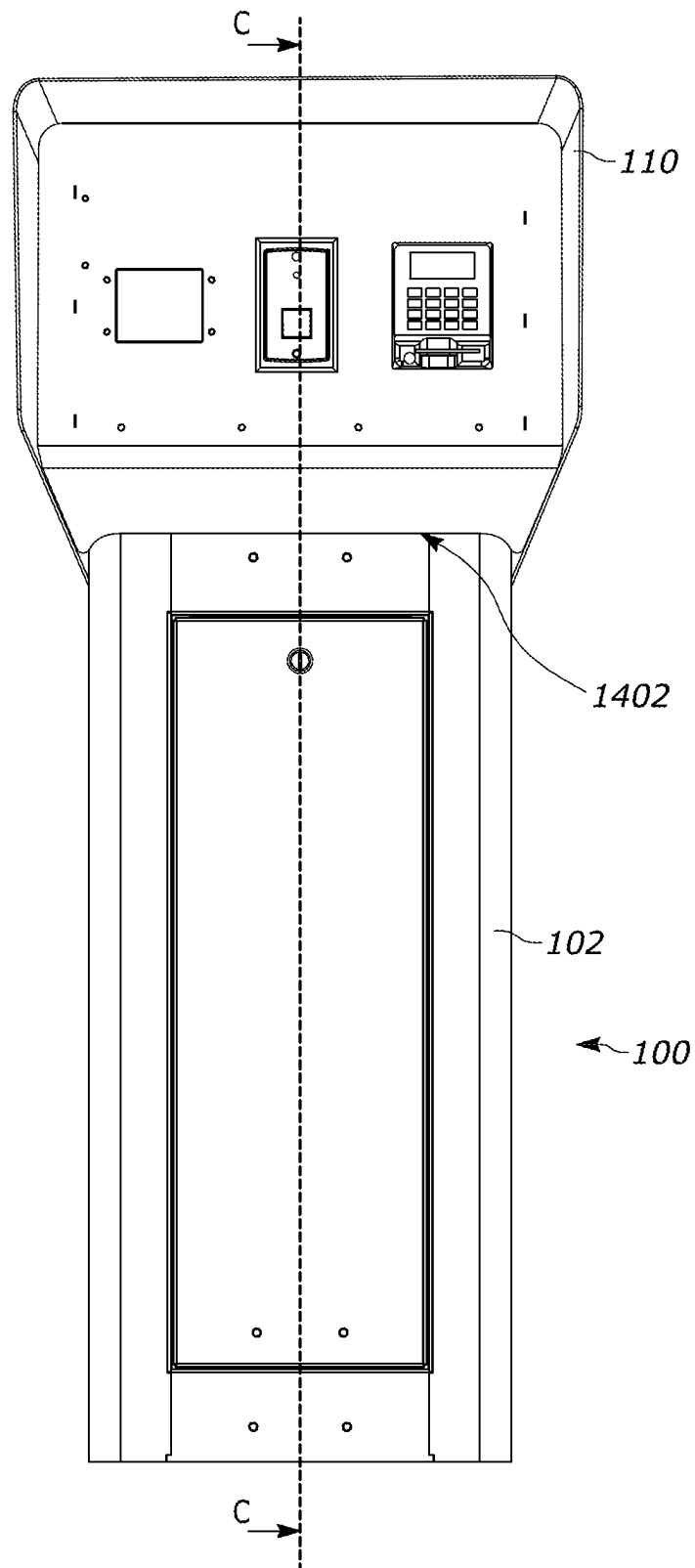


FIG. 14

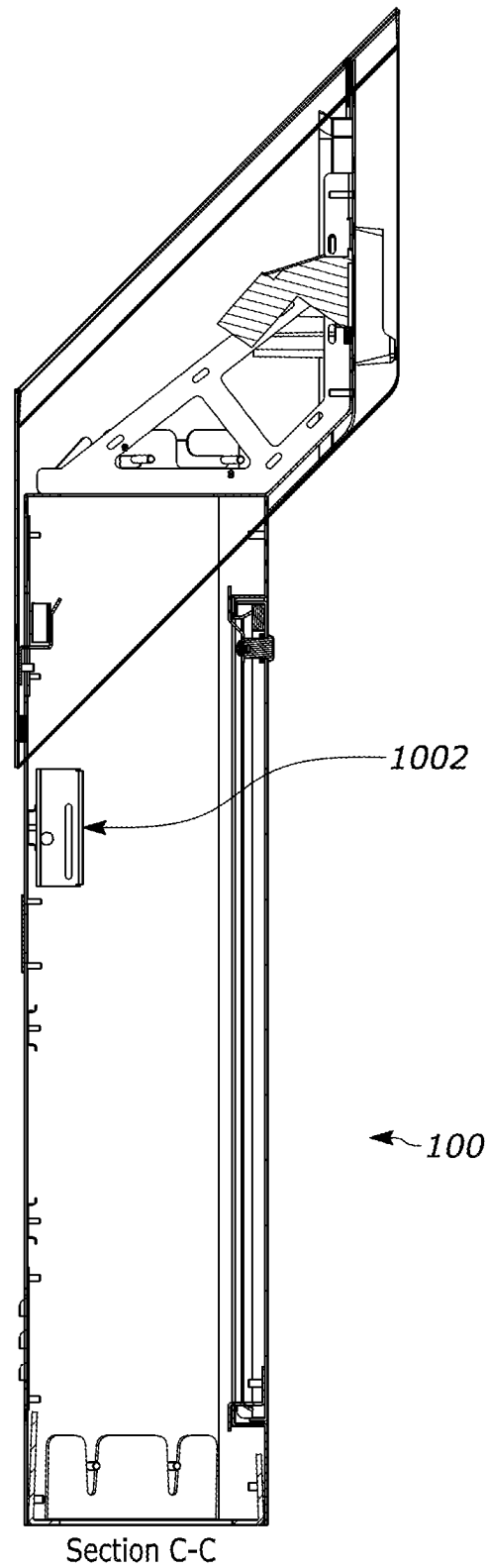


FIG. 15

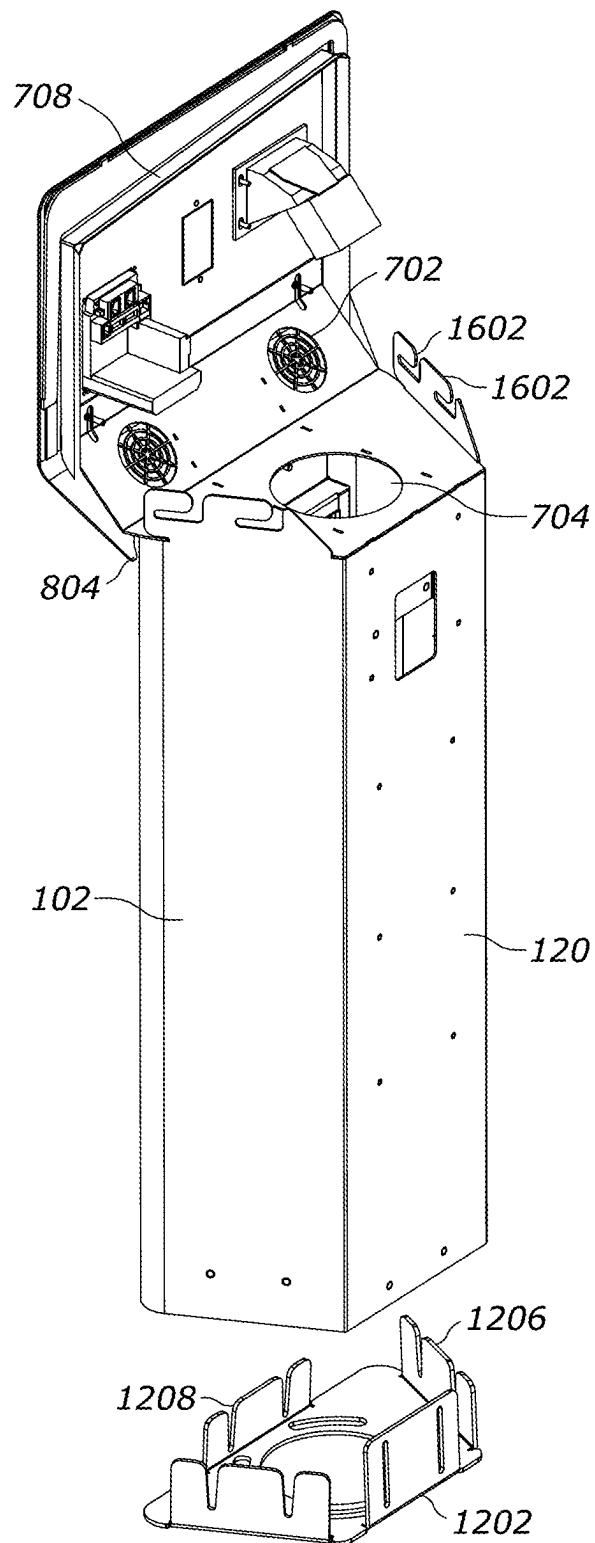


FIG. 16

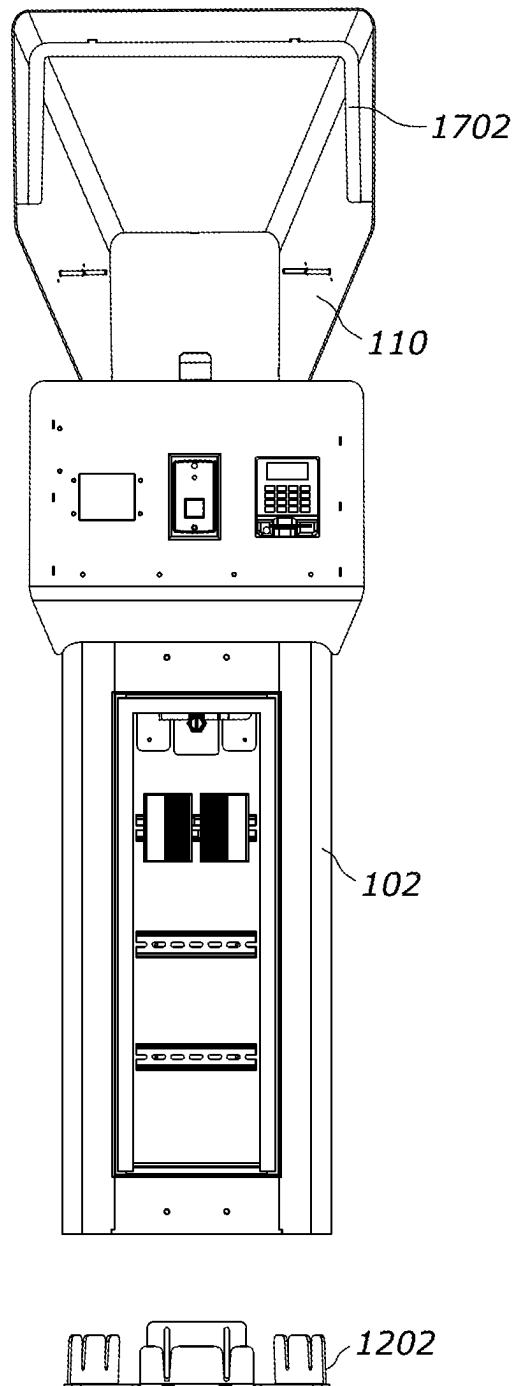


FIG. 17

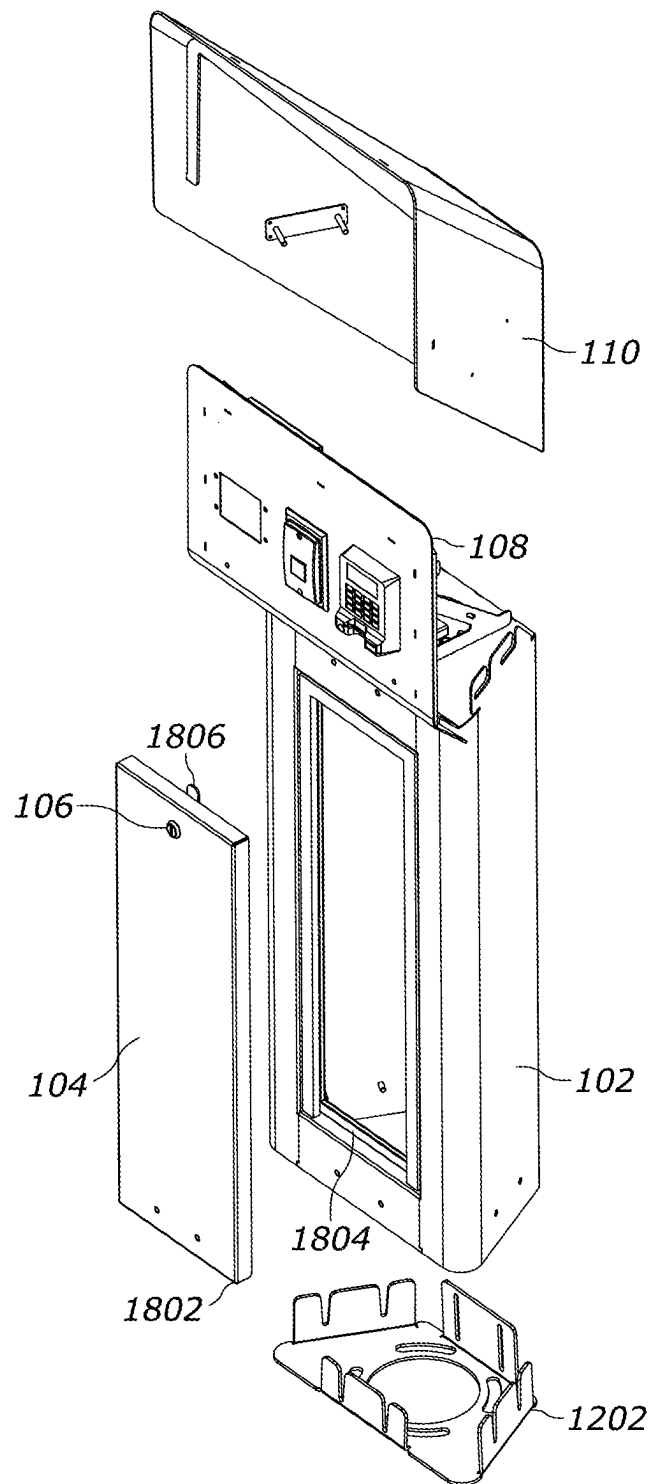
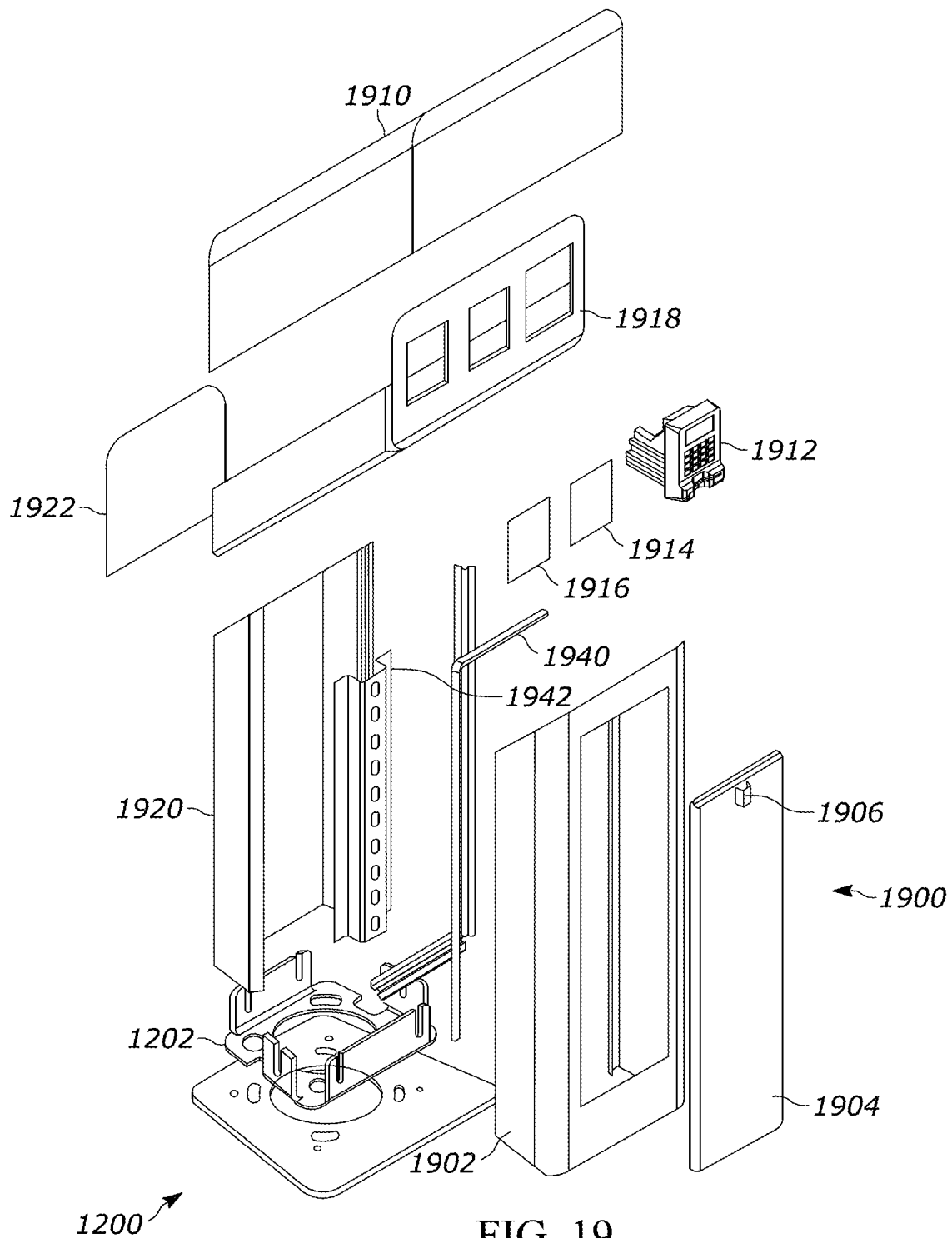


FIG. 18



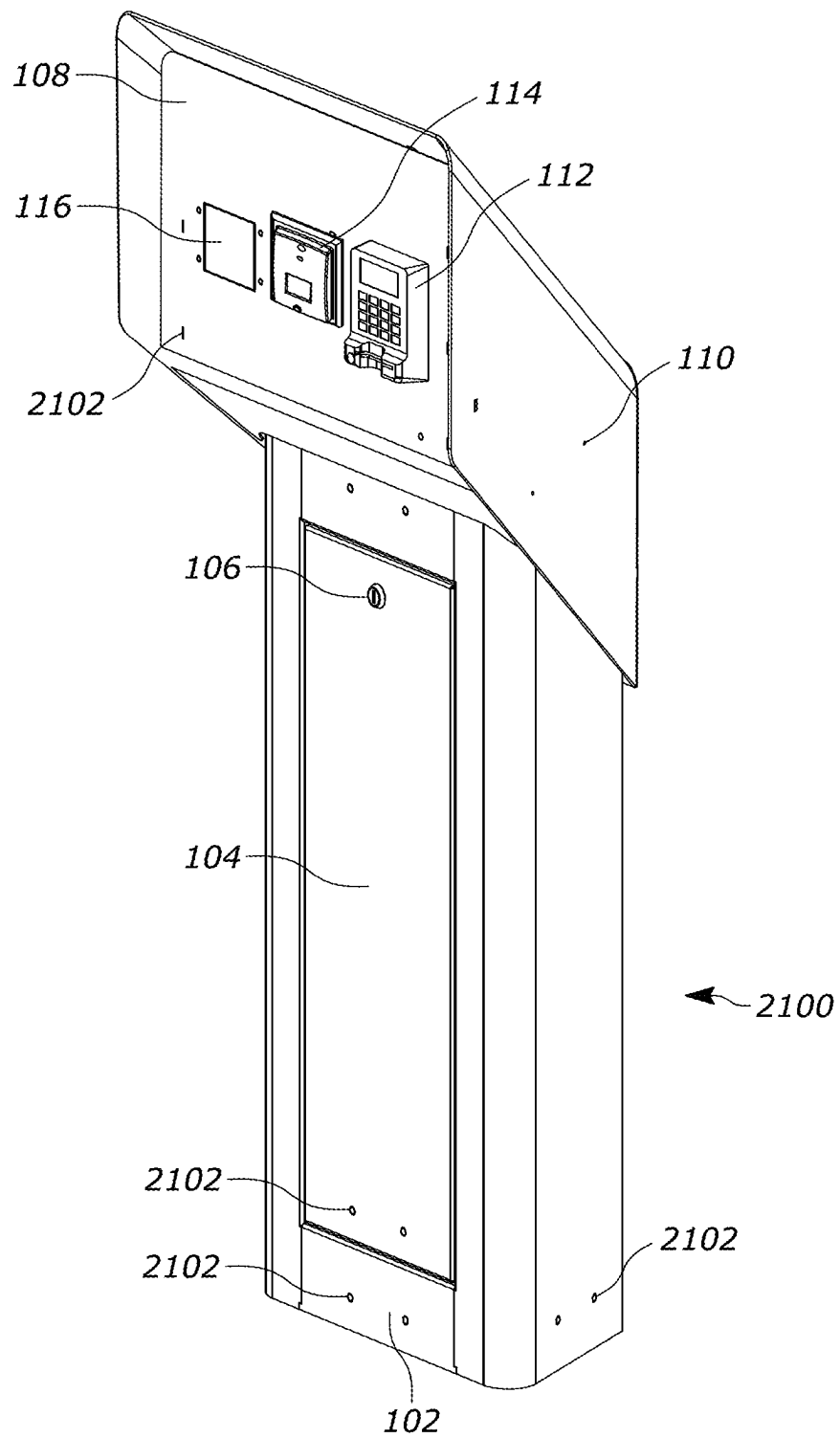


FIG. 20

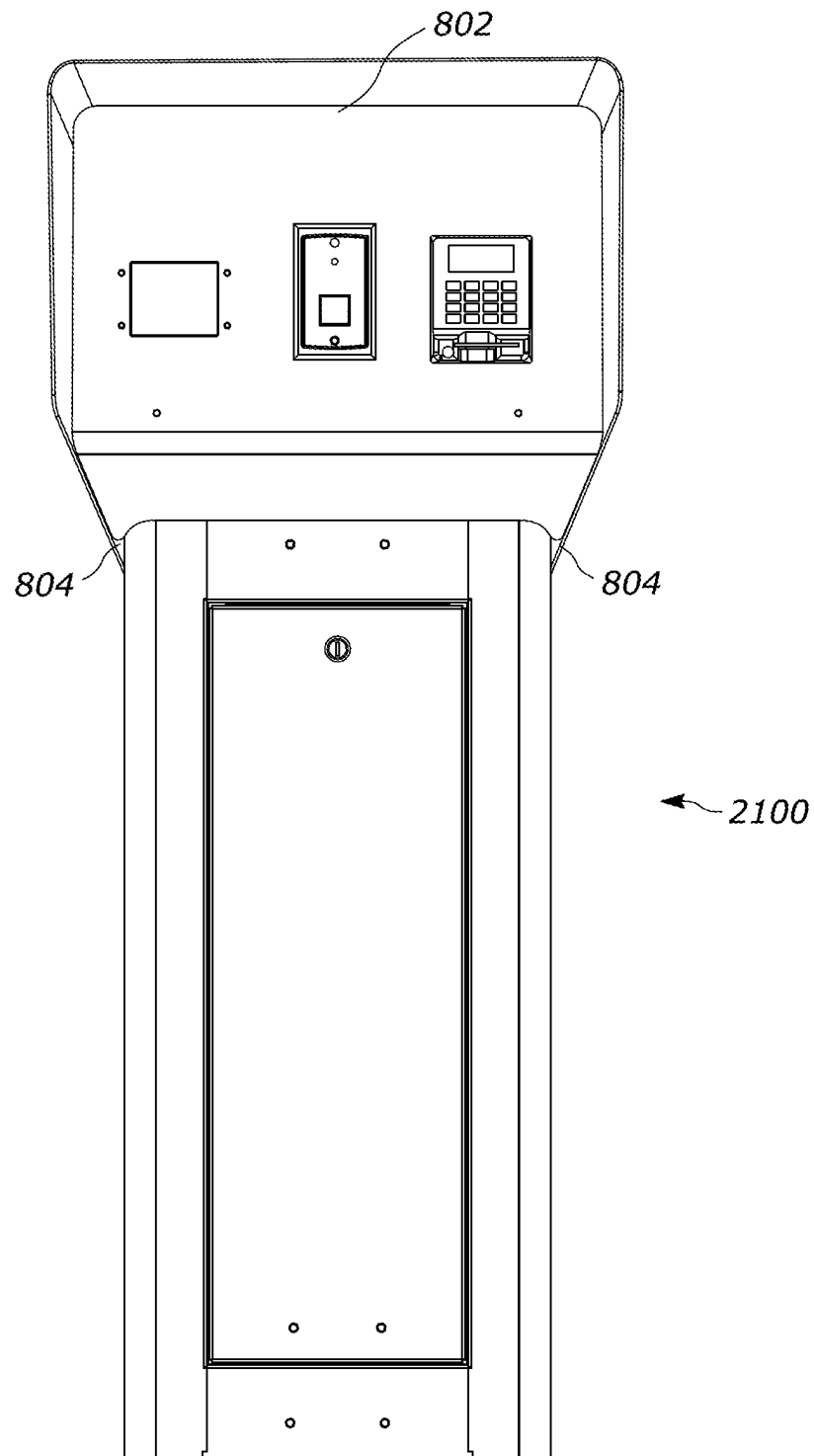


FIG. 21

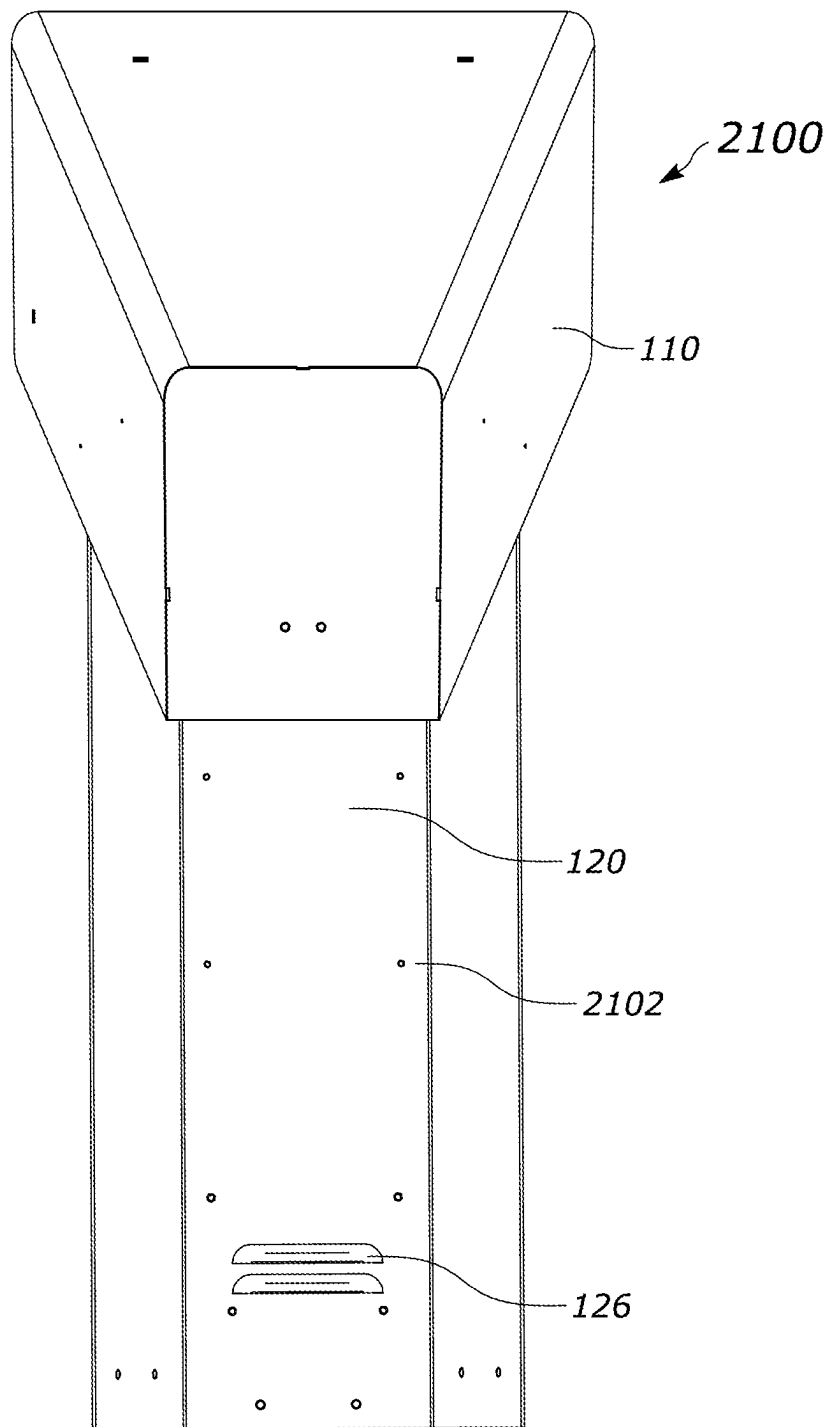


FIG. 22

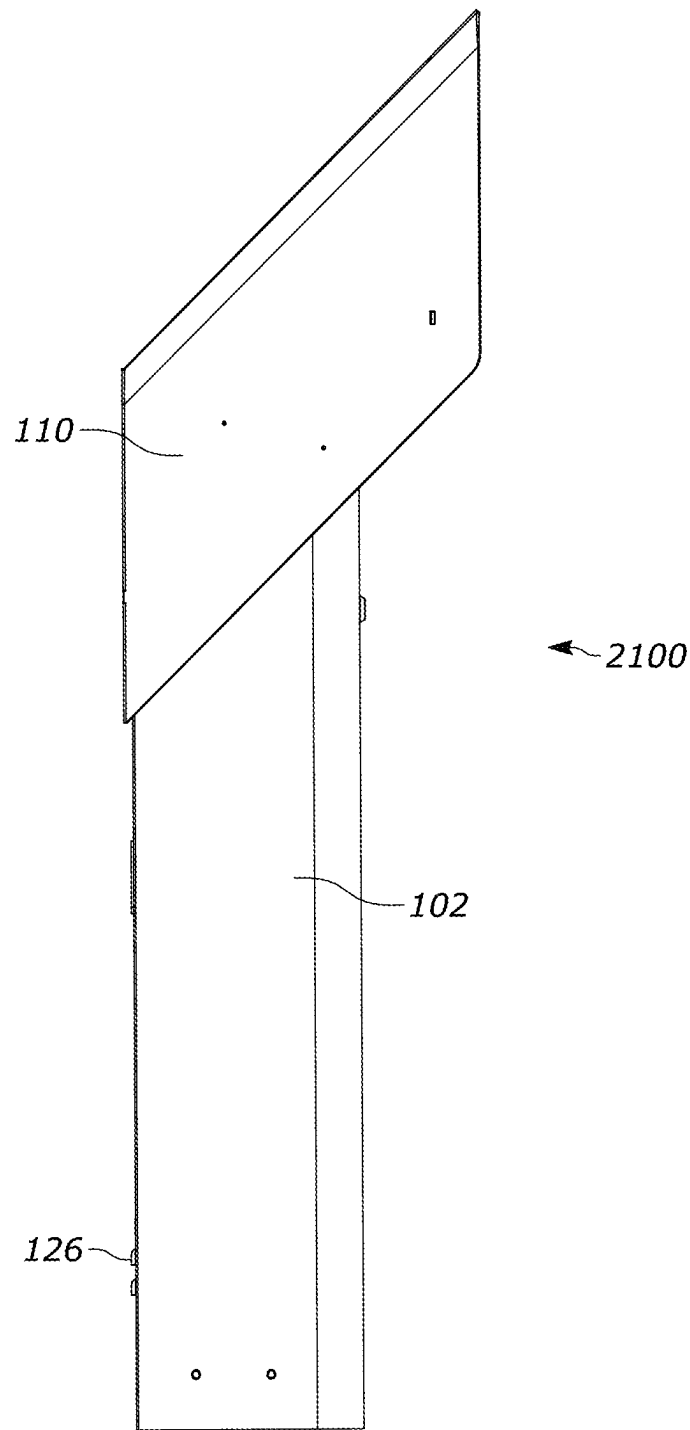


FIG. 23

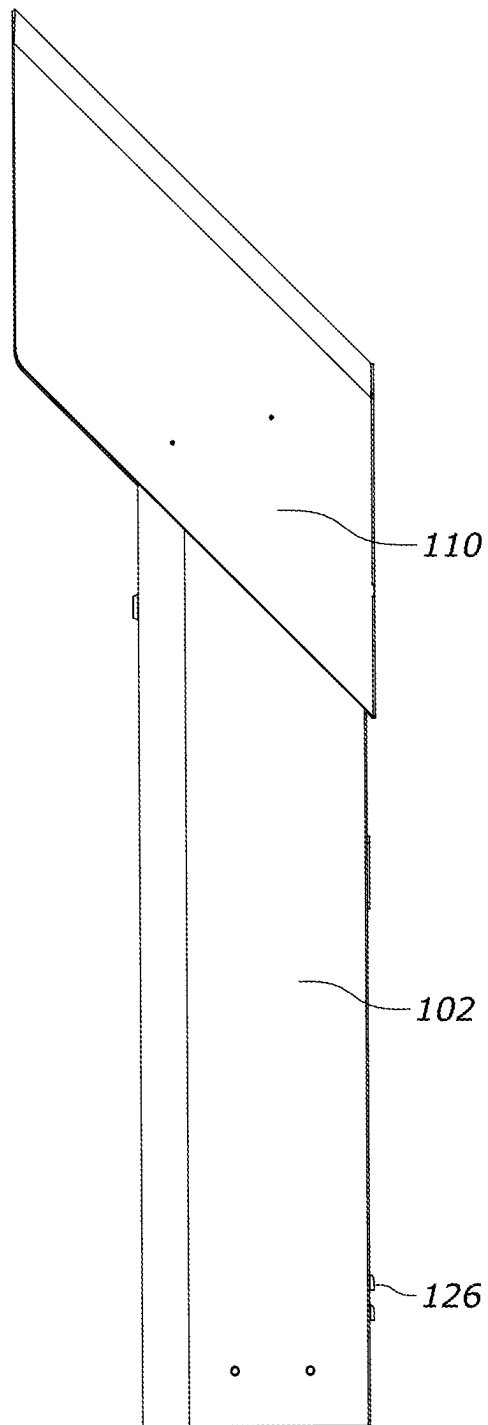


FIG. 24

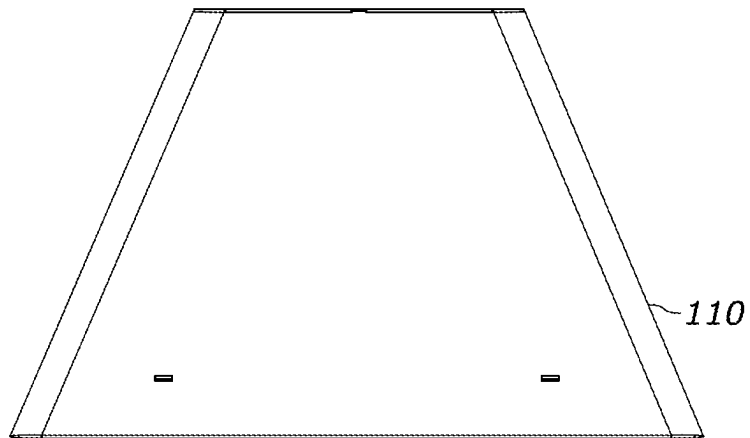


FIG. 25

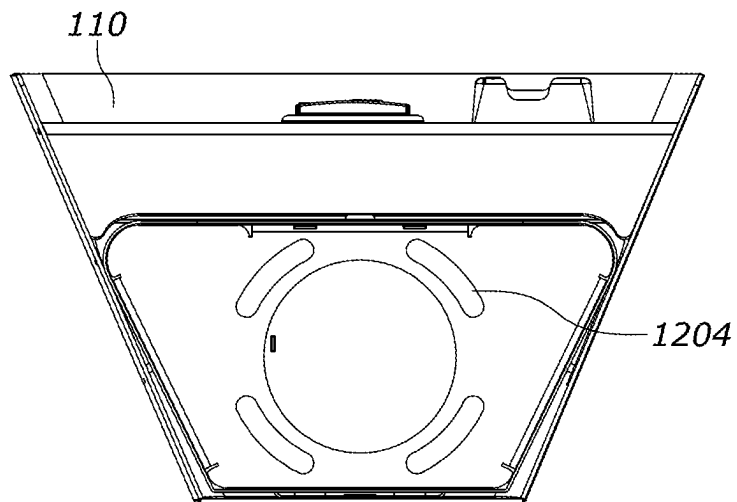


FIG. 26

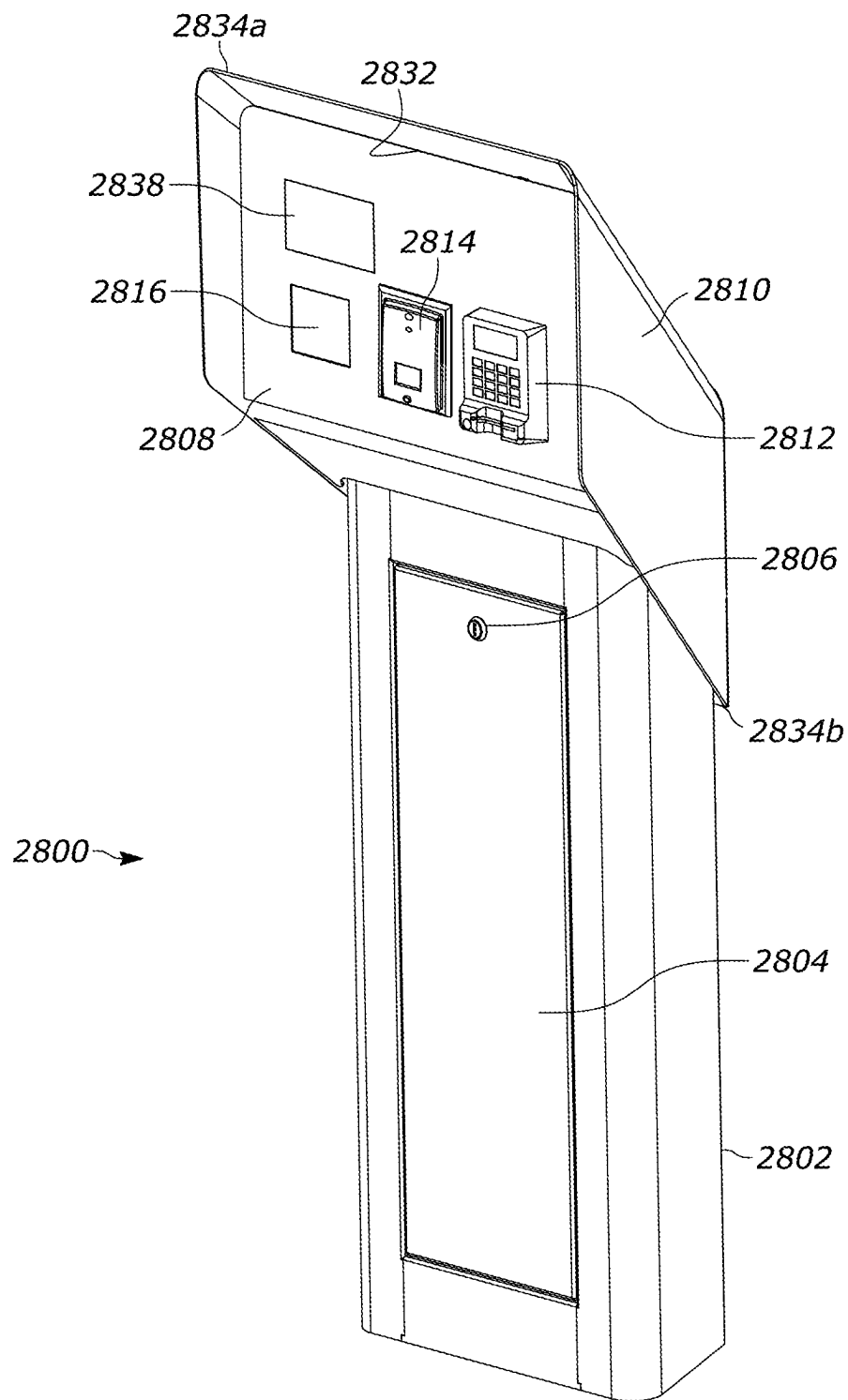


FIG. 27

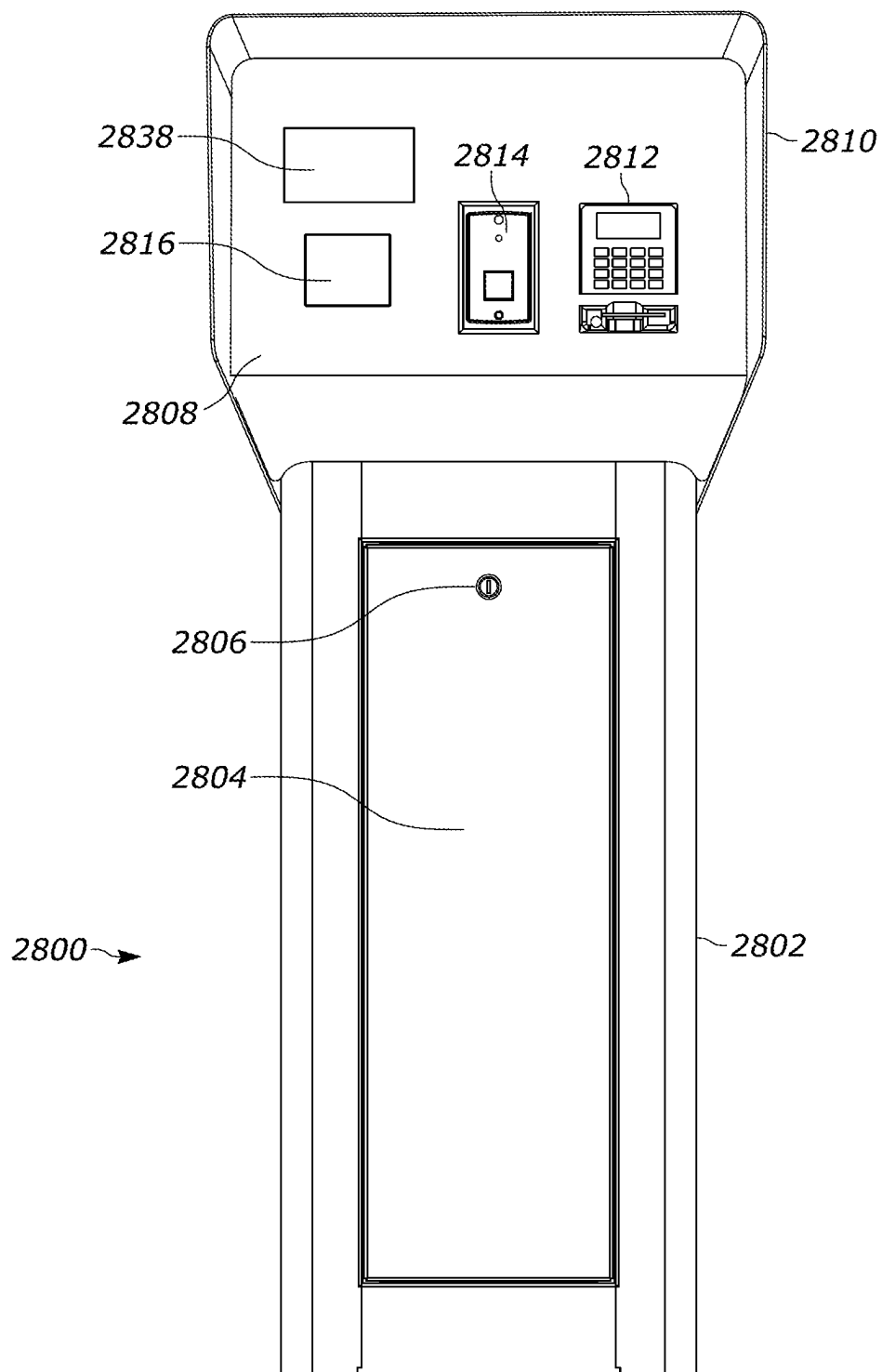


FIG. 28

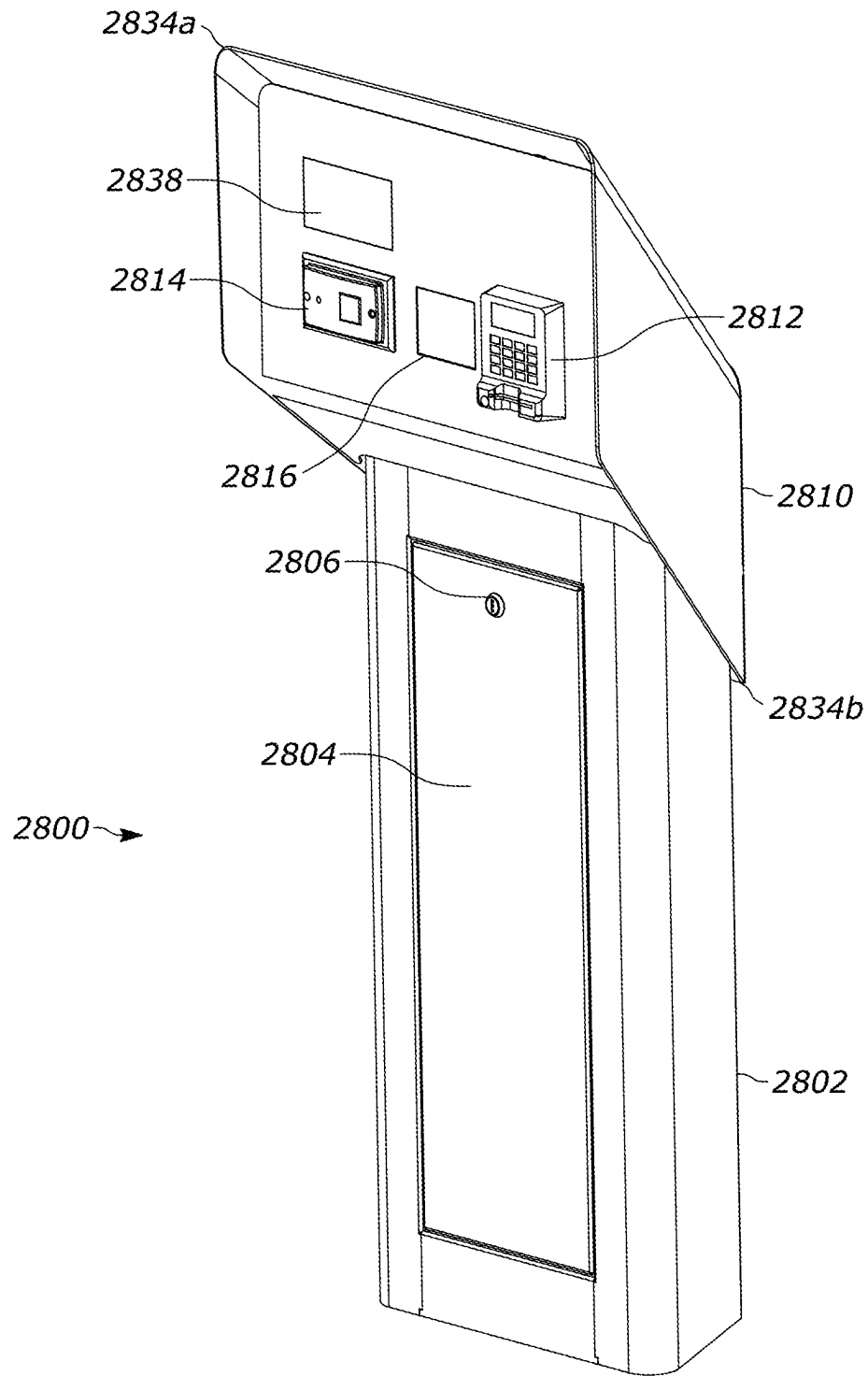


FIG. 29

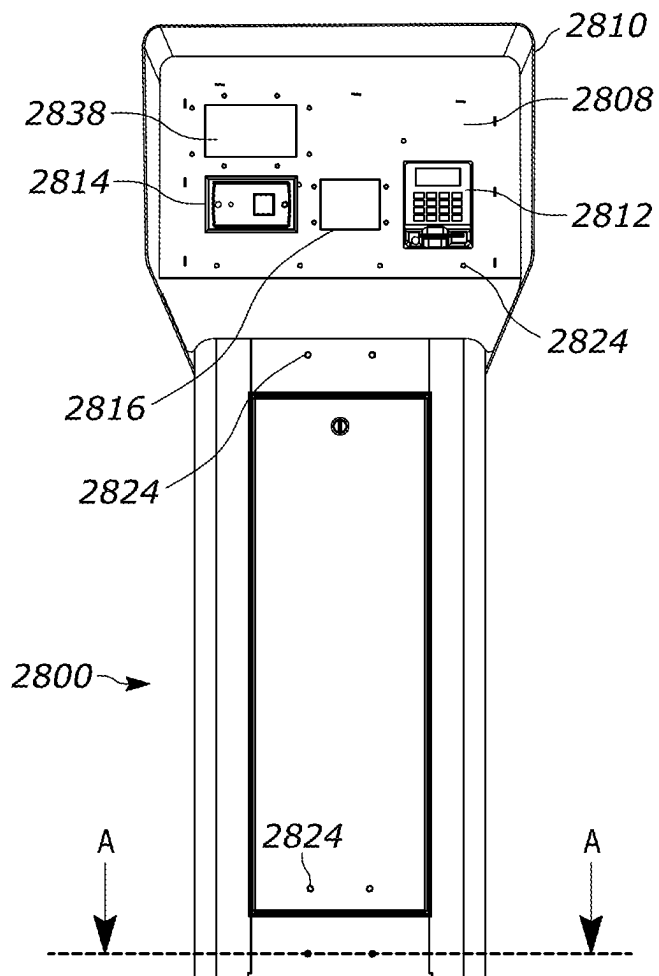


FIG. 30

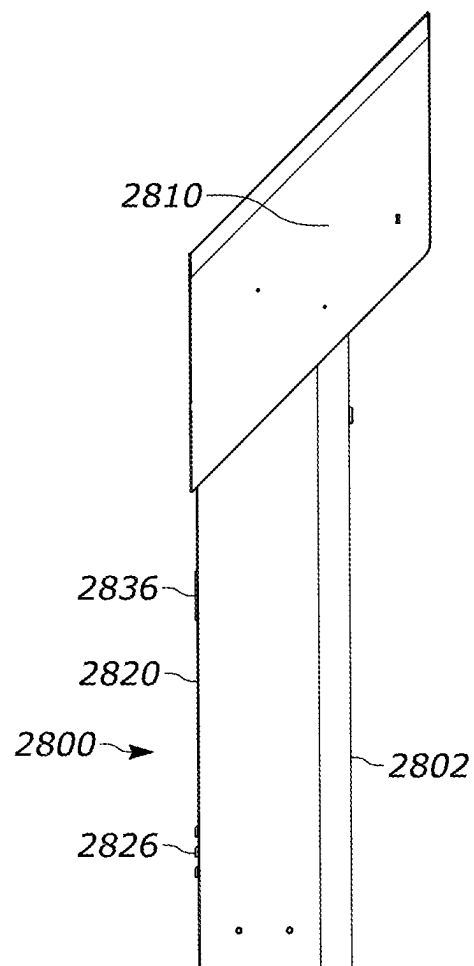


FIG. 31

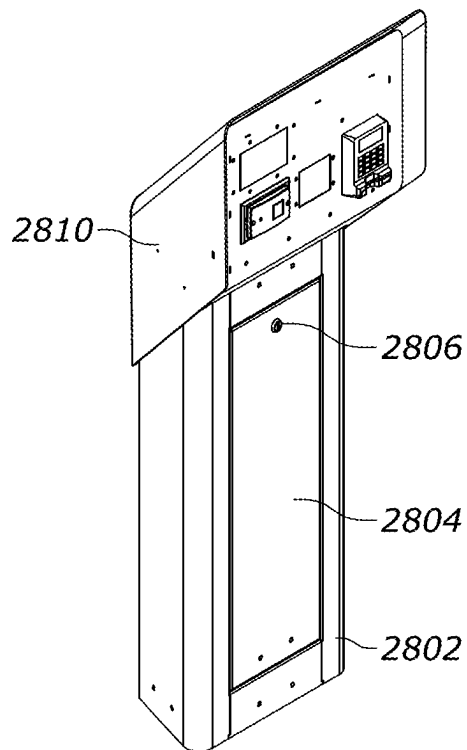


FIG. 32

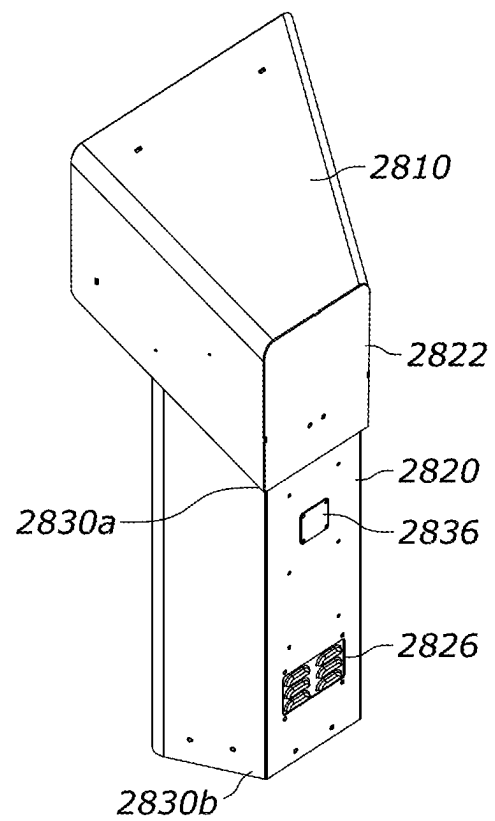


FIG. 33

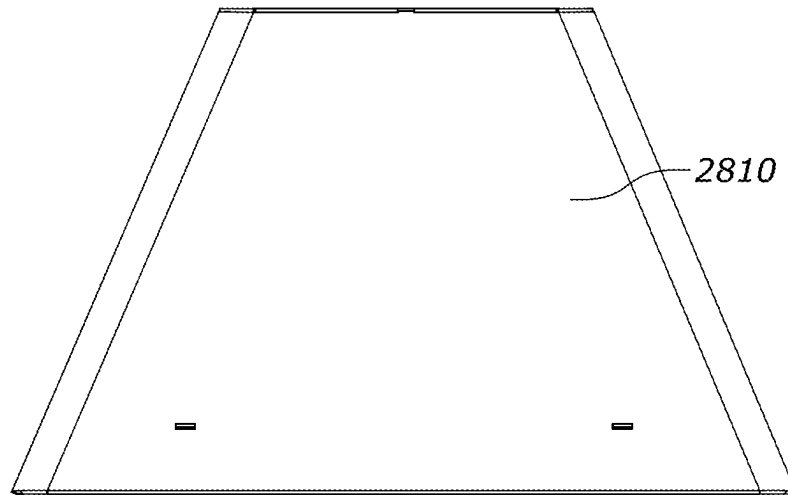
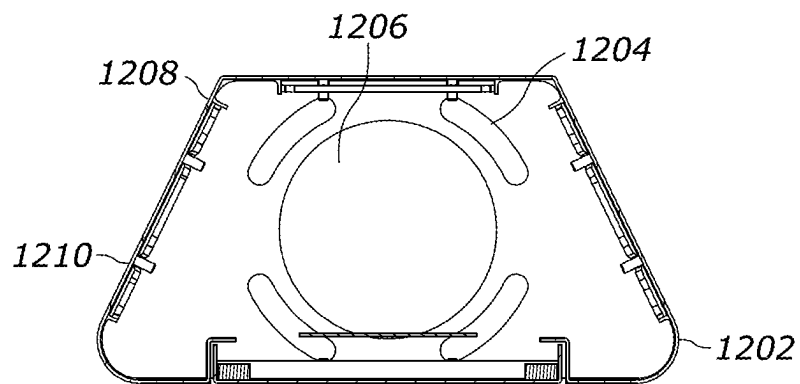


FIG. 34



Section A-A

FIG. 35

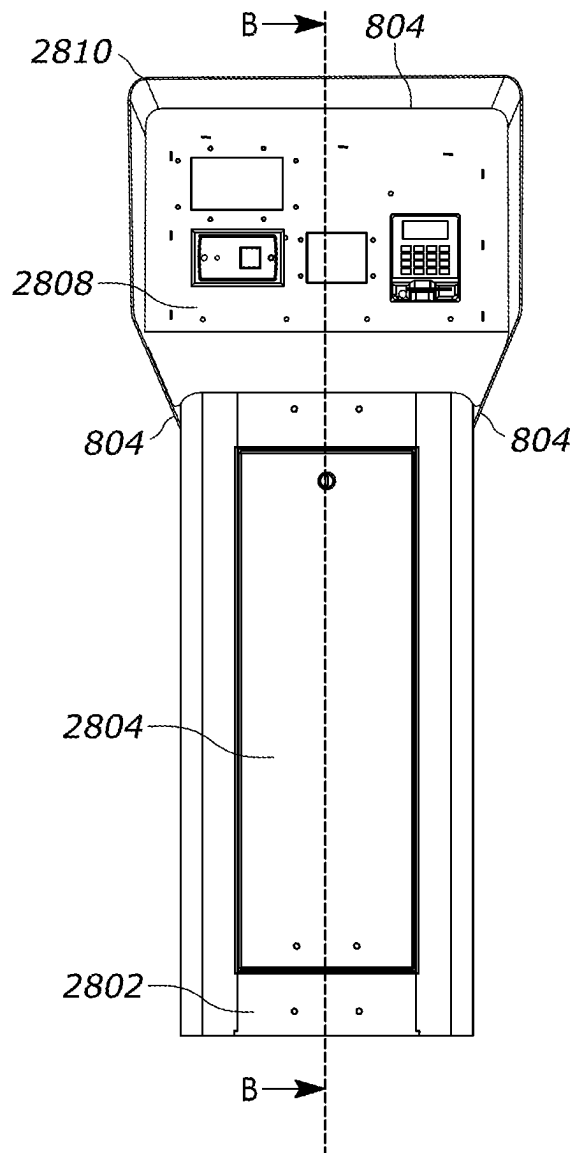


FIG. 36

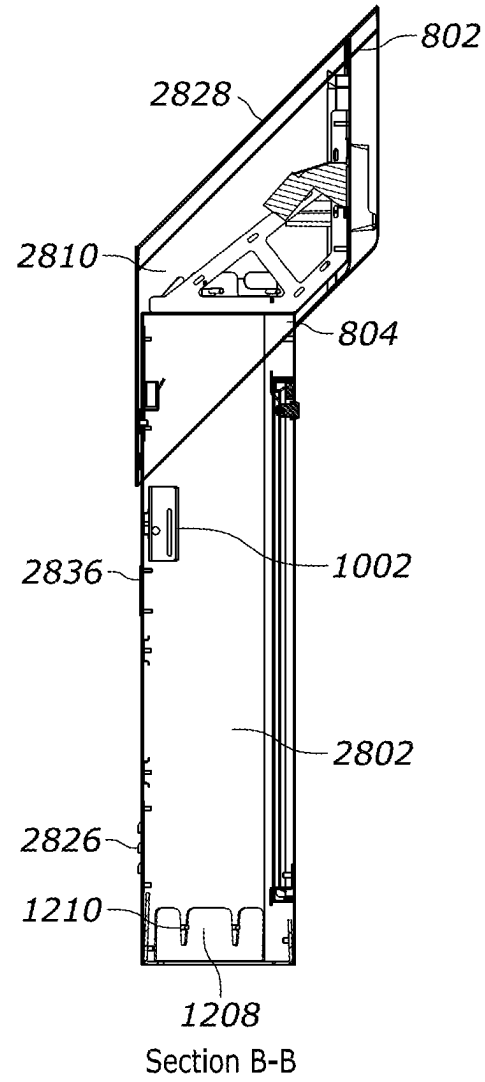


FIG. 37

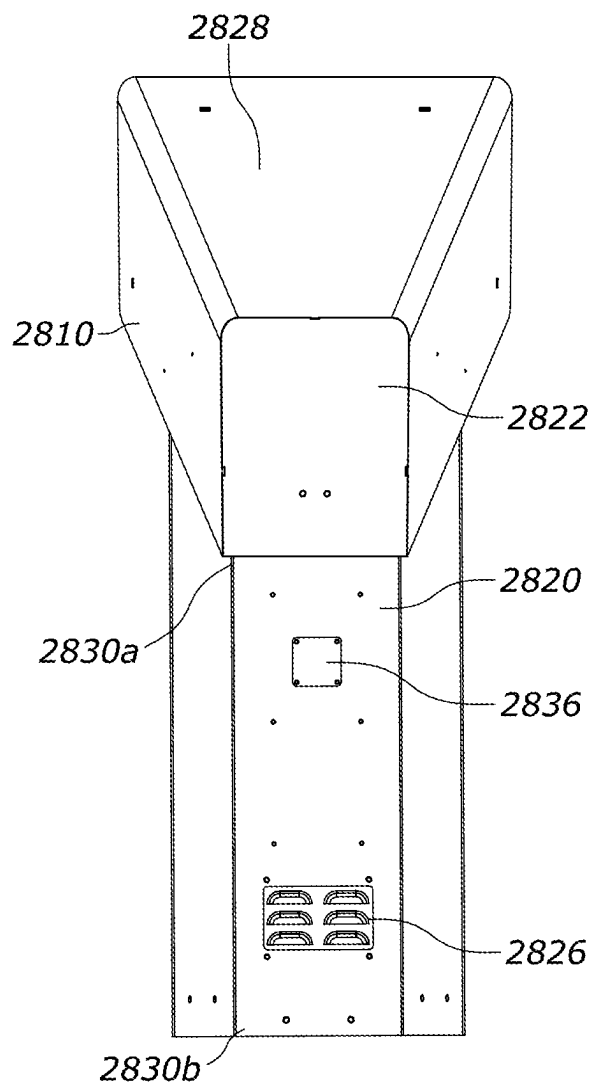


FIG. 38

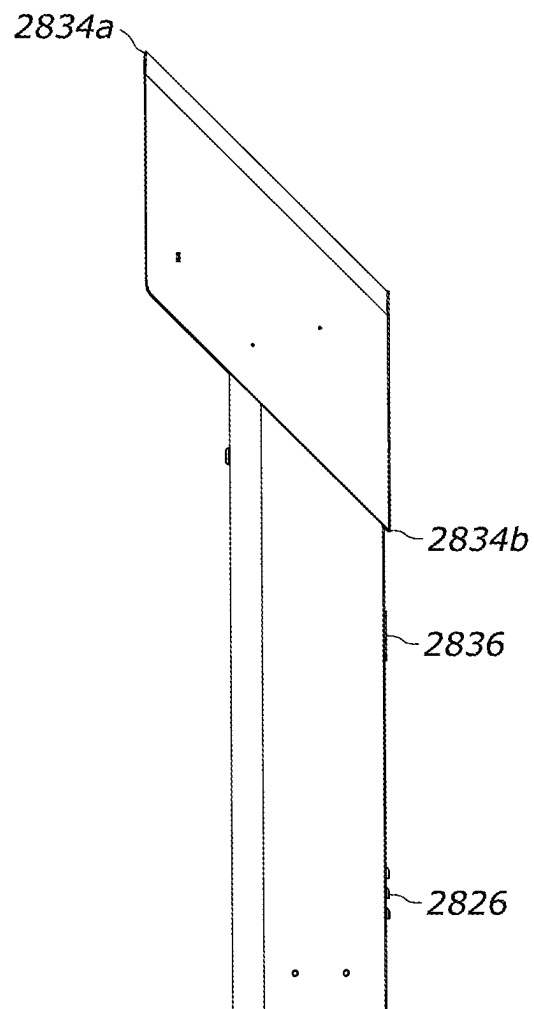


FIG. 39

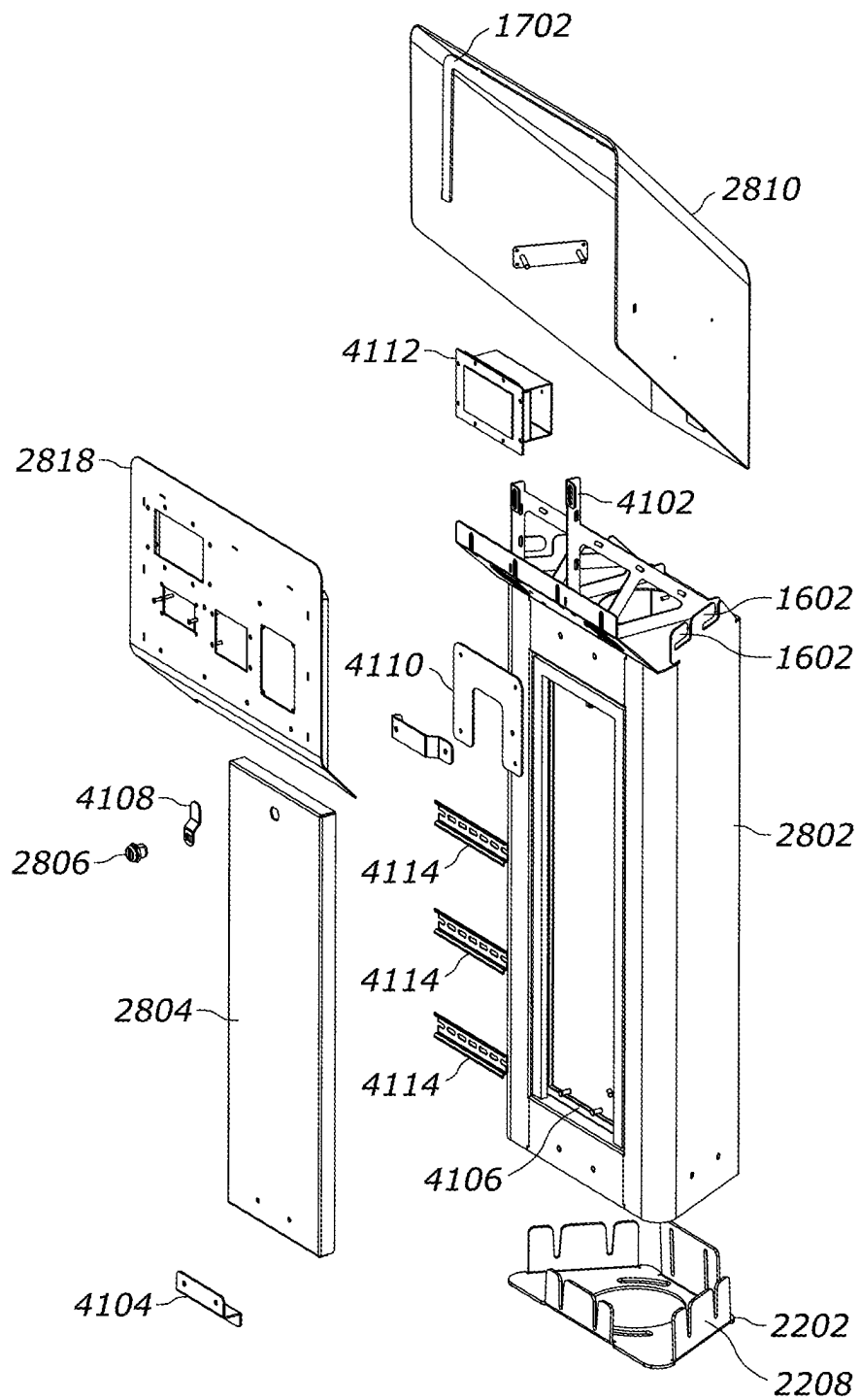


FIG. 40

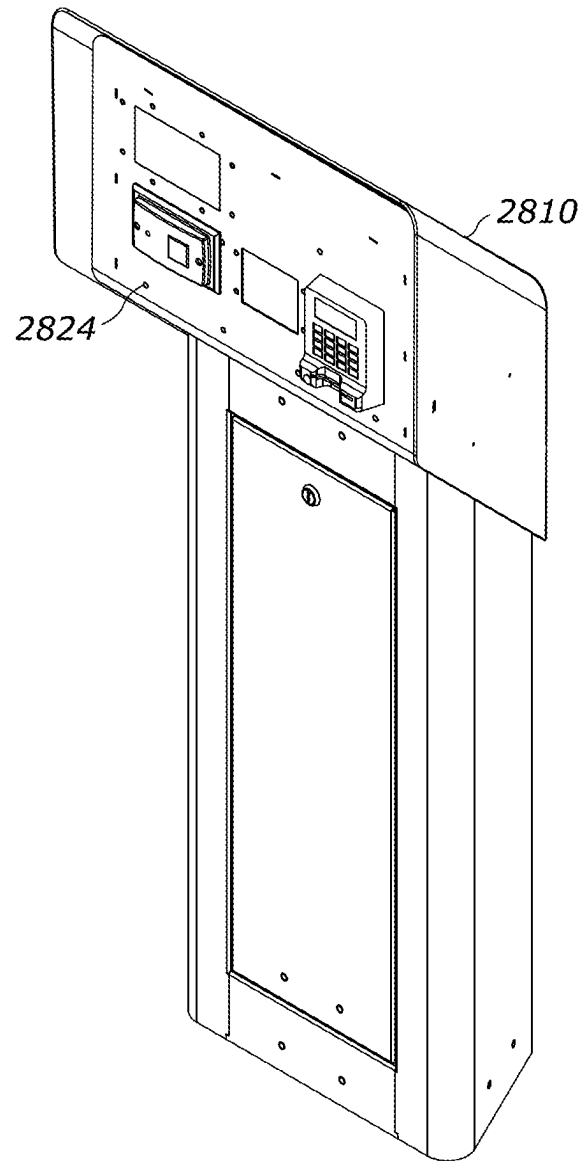


FIG. 41

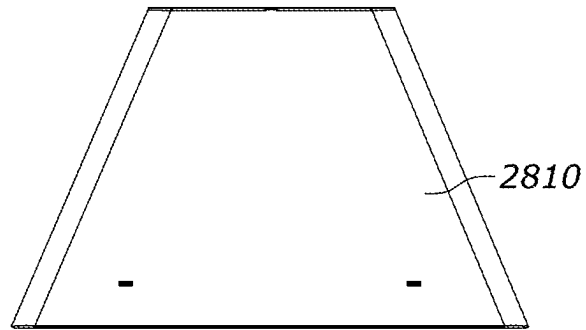


FIG. 42

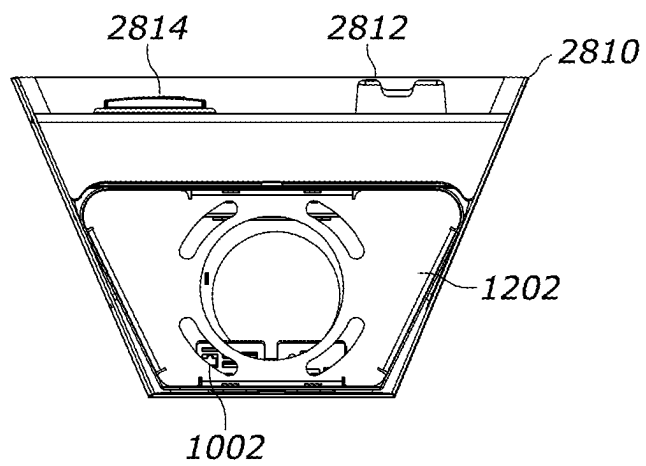


FIG. 43

1

TERMINALS AND METHODS OF USE AND MANUFACTURE THEREOF

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to U.S. Provisional Application No. 63/318,014 filed Mar. 9, 2022, the content of which is hereby incorporated by reference.

COPYRIGHT NOTICE

Copyright 2022 S P Plus Corporation. A portion of the disclosure of this patent document contains material that is subject to copyright protection. The copyright owner has no objection to the facsimile reproduction by anyone of the patent document or the patent disclosure, as it appears in the Patent and Trademark Office patent file or records, but otherwise reserves all copyright rights whatsoever. 37 C.F.R. § 1.71(d).

FIELD OF THE DISCLOSURE

The present disclosure generally relates to systems, methods, and associated devices for use in controlling access to a restricted area. More particularly, the present disclosure relates to a terminal for controlling access by vehicles to a parking area and methods of use and manufacture thereof. In embodiments, a novel terminal is provided.

BRIEF SUMMARY OF THE DISCLOSURE

This summary is provided to introduce a selection of concepts in a simplified form that are further described in the detailed description of the disclosure. This summary is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for determining the scope of the claimed subject matter.

Generally, a parking terminal as disclosed herein may include one or more outer panels and a hood. In embodiments, the hood is sloped and extends above a front face of the terminal.

In embodiments of the present disclosure, a terminal comprises at least one outer panel; a hood located above at least one outer panel; and an inner panel located on a front face of the terminal below the hood, the inner panel comprising one or more reader devices; wherein a top surface of the hood is angled at least 26 degrees from horizontal.

In embodiments, the hood extends past the inner panel on the front face of the terminal. In embodiments, the terminal further comprises a back face with a first vent located on the back face of the at least one outer panel; and a second vent located between the hood and the inner panel. In embodiments, the hood is wider proximate the front face than proximate the back face. In embodiments, a drain is located on the front face between at least one outer panel and the inner panel and wherein a channel is disposed proximate the vent on an interior surface of the inner panel and is angled such that a first end of the channel is lower than an opposite second end of the channel and wherein the first end is closer to the drain than the second end.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

The foregoing summary, as well as the following detailed description of the disclosure, is better understood when read

2

in conjunction with the appended drawings. For the purpose of illustrating the disclosure, exemplary constructions of the inventions of the disclosure are shown in the drawings. However, the disclosure and the inventions herein are not limited to the specific methods and instrumentalities disclosed herein.

FIG. 1 exemplarily illustrates a front view of a first embodiment of a terminal.

FIG. 2 is a side perspective view of the terminal of FIG. 1.

FIG. 3 is a side view of the terminal of FIG. 1.

FIG. 4 is a back view of the terminal of FIG. 1.

FIG. 5A is a front view of a terminal with the internal electronics omitted.

FIG. 5B is a side perspective view of the terminal of FIG. 5.

FIG. 6 is a front view of a second embodiment of a terminal with the outer covering removed.

FIG. 7 is a rear view of the terminal of FIG. 7.

FIG. 8 exemplarily illustrates a front view of a third embodiment of a terminal.

FIG. 9 is a side view of the terminal of FIG. 8.

FIG. 10 is a front perspective view of the terminal of FIG. 8 with the covering and other components shown in broken lines to illustrate internal components.

FIG. 11 is a rear perspective view of the terminal of FIG. 8.

FIG. 12 is a top-down view taken through Section B-B showing the bottom plate of the terminal of FIG. 8.

FIG. 13 is a top view of the terminal of FIG. 8.

FIG. 14 is a second front view of the terminal of FIG. 8.

FIG. 15 is a side view taken through Section C-C of the terminal of FIG. 14.

FIG. 16 is a rear perspective view of the terminal of FIG. 8 with the hood removed and the base plate shown separated from the terminal.

FIG. 17 is an exploded front view of the terminal of FIG. 7 with the access panel omitted.

FIG. 18 is an exploded front perspective view of the terminal of FIG. 9.

FIG. 19 exemplarily illustrates an exploded view of a fourth embodiment of a terminal.

FIG. 20 is a perspective view of a fifth embodiment of a parking terminal.

FIG. 21 is a front view of the terminal of FIG. 20.

FIG. 22 is a rear view of the terminal of FIG. 20.

FIG. 23 is a right side view of the terminal of FIG. 20.

FIG. 24 is a left side view of the terminal of FIG. 20.

FIG. 25 is a top view of the terminal of FIG. 20.

FIG. 26 is a bottom view of the terminal of FIG. 20.

FIG. 27 is a perspective view of a sixth embodiment of a parking terminal.

FIG. 28 is a front view of the parking terminal of FIG. 27.

FIG. 29 is a second perspective view of the sixth embodiment of a parking terminal with the devices rearranged on the inner panel.

FIG. 30 is a front view of the parking terminal of FIG. 29.

FIG. 31 is a side view of the parking terminal of FIG. 29.

FIG. 32 is a front perspective view of the parking terminal of FIG. 29.

FIG. 33 is a rear perspective view of the parking terminal of FIG. 29.

FIG. 34 is a top view of the parking terminal of FIG. 29.

FIG. 35 is a bottom view of the parking terminal of FIG. 29.

FIG. 36 is a second front view of the parking terminal of FIG. 29.

3

FIG. 37 is a cutaway side view of the parking terminal of FIG. 29.

FIG. 38 is a rear view of the parking terminal of FIG. 29.

FIG. 39 is a second side view of the parking terminal of FIG. 29.

FIG. 40 is an exploded view of the parking terminal of FIG. 29.

FIG. 41 is a third perspective view of the parking terminal of FIG. 29.

FIG. 42 is an additional top view of the parking terminal of FIG. 29.

FIG. 43 is an alternative bottom view of the parking terminal of FIG. 29.

DETAILED DESCRIPTION OF THE DISCLOSURE

The following disclosure as a whole may be best understood by reference to the provided detailed description when read in conjunction with the accompanying drawings, drawing description, abstract, background, field of the disclosure, and associated headings. Identical reference numerals when found on different figures identify the same elements or a functionally equivalent element. The elements listed in the abstract are not referenced but nevertheless refer by association to the elements of the detailed description and associated disclosure.

Express Terminal

As shown in FIGS. 1 through 43, embodiments of the present disclosure provide a terminal 100, 1900, 2100, and 2800 (also referred to herein as an Express Terminal or ET) for use in controlling access to an area. In embodiments, the terminal manages access to a parking facility. The terminal 100 generally acts cooperatively with an access control mechanism—such as a gate, door, or other physical barrier that physically impedes access to a facility or a virtual access control or other barrier or enforcement mechanism such as License Plate Recognition, License Plate Enforcement, or another enforcement such as towing of unauthorized vehicles—to ensure that only authorized users are permitted vehicular access to the facility, only authorized users (i.e., users who have paid) are permitted to depart a facility, and/or that unauthorized vehicles within a facility are identified so that remedial and/or enforcement action may be taken. The parking facility may be, by way of example, a parking garage, surface lot, or other restricted area into which only authorized vehicles are permitted access. In embodiments, the terminal 100 is located proximate an access control mechanism at the entrance and/or exit to a facility. In embodiments, the terminal 100 is connected to a broader access control system (as discussed in greater detail below) and serves as a physical interaction point for a user desiring to enter or leave the parking facility. In other embodiments, the terminal functions as a stand-alone device that works directly with the access control mechanism and a payment processor to control access to the parking facility.

As shown, in embodiments, the terminal 100 comprises a front plate 102, access panel 104, hood 110, inner panel 108, back plate 120, and upper back plate 122 that cooperatively form the exterior surface of the terminal 100. As will be clear to one of skill in the art, in embodiments, the foregoing components may each comprise multiple elements or may be combined into single physical structures. By way of example, in an embodiment, the front plate 102 comprises multiple plates that are separately formed before being jointed together while the back plate 120 and upper back plate 122 comprises a single unitary structure formed as a

4

single piece. Other alternative arrangements will be clear to one of skill in the art from the present disclosure.

One or more devices may be disposed along inner panel 108. In the embodiment shown in FIG. 1, a scanner/reader 116 for reading the barcode of a card, paper, or mobile device; an RFID reader 114 for optically reading a radio-frequency ID (RFID) and or near field communication (NFC); credit card reader 112, and PIN pad 112 are each disposed on the inner panel 108. As will be clear to one of skill in the art, other arrangements and/or combinations of devices may also or instead be disposed on the inner panel 108. Each of the devices 112, 114, and 116 is coupled to a controller 1002 and power supply (not shown), which may be provided through electrical mains and/or a battery, such as through wiring 706. A central channel 704 in the terminal 100 permits the routing of cabling 706 between components as well as the flow of air through the terminal 100 (as discussed in greater detail below). In embodiments, one or more speakers 702 and/or one or more microphones (not shown) are also disposed on the inner panel 108. In embodiments, channels 1602 are configured to receive corresponding protrusions on the hood 110 so as to enable the hood to be removably connected to the body of the terminal 100.

Advantageous aspects of terminals 100 in accordance with the present disclosure are discussed separately below. Ease of Installation and Maintenance

Embodiments of terminals 100 are designed so as to permit ready access to the interior of the terminal 100 and permit ready installation of the terminal 100 with limited or no use of specialized tools.

Embodiments provide an easily movable or removable access panel 104, whereby the access panel 104 is movably or removably connected to the front plate 102 such that the access panel 104 may be moved or removed in order to gain access to the interior of the terminal 104. In the embodiment shown in FIG. 18, the front panel 104 comprises a raised lip 1802 on a lower edge thereof that interlocks with a corresponding channel 1804 on a lower interior edge of the front plate 102. A locking protrusion 1806 is movably connected to the keyhole 106 such that the locking protrusion 1806 rotates as the keyhole 106 is turned. With the access panel 104 inserted in the front plate 102 and the locking protrusion 1806 in an upwards-extending position, the access panel 104 is secured to the front plate 102 and cannot be removed. To remove the access panel 104, the keyhole is actuated and rotated, causing the locking protrusion 1806 to rotate and disengage from the rim of the front panel 102. The access panel 104 can then be tilted away from the terminal 100 and entirely removed, leaving an opening in the front panel 102 through which the interior of the terminal 100 may be accessed. This ensures the interior of the terminal 100 cannot be accessed without actuating the keyhole 106 (i.e., by using the corresponding key) and that technicians can readily access the interior of the terminal 100.

In alternative embodiments, no hinges are used to movably connect the access panel 104 to the front plate 102; the lower portion of the access panel is secured by a lock channel to allow for easy removal of the panel completely. Other structures for providing a movable or removable access panel will be clear to one of skill in the art in view of the instant disclosure and are also expressly contemplated hereby.

As shown, the covering of the terminal 100 may be formed from multiple structural pieces that are bent, utilizing few if any welds and/or connectors. In the embodiment shown in FIG. 1, the front plate 102 and back plate 120 are each formed from a single piece of stamped metal that is

5

bent into the desired shape before and therefor fit together. Similarly, the hood **110** is formed from a single piece of stamped metal before the inner baffle **1702** is welded to the hood, as shown in FIG. 17. In this manner, manufacturing costs and complexity may be greatly reduced.

In embodiments, the terminal **100** comprises a removable bottom plate **1202** on which the entire terminal **100** rests. Such configuration greatly improves the ease of installation of a terminal **100**. In use, the bottom plate **1202** is positioned at a desired location and leveled. Inner opening **1206** may be located so as to allow cabling (not shown) to enter the terminal **100**, including but not limited to pre-installed cabling to provide mains power and/or communication connections to the terminal **100**. Outer openings **1204** are configured to permit fasteners (such as screws or bolts) to be driven through the bottom plate **1202** to secure the bottom plate **1202** in place. In embodiments, openings **1204** are curved so as to allow the bottom plate **1202** to be turned (e.g., to avoid rebar or other embedded obstacles within the substructure) and are extended so as to allow fasteners to be placed at preferred locations. The raised edge **1208** of the plate **1202** and notches **1210** serve to cooperatively receive the bottom edge **130b** of the back plate **120** and/or front plate **102** so as to allow the terminal to be secured in place. In embodiments, fasteners may be used to further secure the terminal **100** to the bottom plate **1202**.

The removable bottom plate **1202** beneficially enables the bottom plate **1202** to be easily located and levelled without the need to lift, move, and/or adjust the larger terminal **100** as a whole. Further, the ability to connect the terminal **100** to the bottom plate **1202** without tools enables faster installation.

Passive Cooling

In embodiments, the terminal **100** is passively cooled (i.e., does not incorporate any fans or other active cooling systems). Such configuration reduces the number of components (and thus component cost, installation complexity, and points of failure for the terminal) as well as the operating cost of the terminal (e.g., by reducing ongoing power requirements by eliminating the need to expend energy cooling the terminal **100**). Embodiments contemplated hereby incorporate one or more of the following features.

As shown in FIG. 9, in embodiments, the top surface **128** of the hood **100** is angled. As will be clear to one of skill in the art, the top surface **128** is the portion of the terminal **100** that receives the greatest amount of solar radiation (and thus, solar heating). Angling the top surface **128** serves both to aid in expelling water from the top surface **128** prior to its evaporation (thereby keeping the top surface **128** cleaner and ensuring puddles of water do not form and gain entrance into the terminal **100**) and in minimizing the thermal effect of solar radiation striking the top surface **128**. The angle of the "tilt" (α) of the top surface **128** may be determined based on the intended geographic location for installation of the terminal **100**. By way of example, a terminal **100** intended for installation in Ottawa, Canada may utilize a lower degree of tilt than a terminal intended for installation in Miami, Fla. based on the lesser incidence of solar radiation at more northern latitudes. In embodiments, the tilt is greater than 26 degrees from the horizontal. In alternative embodiments, the tilt is between approximately 40 and 50 degrees from horizontal. In the embodiment shown in FIG. 18, the top surface is angled 135 degrees from vertical (i.e., it is tilted "upwards" 45 degrees above horizontal).

In contrast, in embodiments, the front plate **102** and upper back plate **122** are substantially vertical, thereby minimizing the area receiving solar radiation on the surfaces thereof.

6

As shown in FIG. 1, in embodiments the hood **110** extends over the inner panel **108** such that the hood **110** provides a degree of shading to the inner panel **108**. In embodiments, that overhang may be increased (e.g., to provide greater shading to the inner panel **108**).

To further aid in dissipating solar radiation, the exterior surface of the terminal **100** may be a light or reflective color, such that solar radiation is reflected away from the terminal **100**. In the embodiment shown in FIG. 18, the hood **110** (including top surface **128**) and front panel **102** are painted or otherwise colored to the RAL-9010 shade of white while the inner panel **108** is painted or otherwise colored to the RAL-9006 shade of aluminum grey.

In embodiments, vents **126** are provided in the back plate **120**. In the embodiment shown in FIG. 4, a first set of vents **126** are disposed proximate the top edge **130a** of the back plate **120** while a second set of vents **126** are disposed proximate a lower edge **130b** of the back plate **120**. The first set of vents **126** is thereby located above the controller **1002** while the second set of vents **126** is disposed below the controller **1002**. In the embodiment shown, both sets of vents **126** serve as air intakes, allowing ambient air to enter the terminal **100** therethrough. As the ambient air is warmed (e.g., through waste heat generated by the electrical components disposed within the terminal and/or solar radiation impacting the exterior surface of the terminal **100**), it rises through the interior of the terminal **100** before being expelled through a top vent **802** disposed between the top edge **132** of the interior panel **104** and the hood **110**, as shown in FIG. 11. Vents **126**, **802** are comprised of openings to permit the passage of air.

The hood **110** is angled such that a front edge **134a** is wider than a back edge **134b**. In the preferred embodiment, the angle between the sides of the hood **110** (α) is between approximately 40 and 50 degrees. This aids in creating a chimney effect whereby air exiting through the top vent **802** creates a lower pressure zone that accelerates the flow of air through the terminal **100** (and along the surfaces of the electrical components disposed therein), increasing the effective cooling rate. In embodiments, the top vent **802** extends along the entire edge **132** of the interior panel **108**. In alternative embodiments, one or more top vents **802** are located along a portion of the edge **132**.

Water Redistribution

In embodiments, the exterior of the terminal **100** is made from a waterproof material (such as metal or plastic). In further embodiments, the terminal **100** is specifically designed to ensure water entering the terminal **100** is expelled without accessing electrical or other sensitive components. As shown in FIG. 7, in an embodiment, a back channel **708** is angled across the interior surface of the inner panel **108** and protrudes horizontally towards the back of the terminal **100**. The back channel **708** functions as a gutter, catching water that gains entrance to the terminal **100** around the edges of the inner panel **108** and directing it down towards a lower corner of the inner panel **108** to drain out of the terminal **100**. In embodiments, water may enter through a top opening **802** and is expelled through a bottom opening **804** disposed between the inner panel **108** and hood **110**.

Fourth Embodiment

FIG. 19 illustrates an exploded view of a fourth embodiment of a terminal **1900** that is generally similar to terminal **100** except as otherwise noted herein.

As shown, the terminal comprises a base **1200** that is joined to a base plate **1202** during installation. The base **1200** is placed at a desired location (e.g., such that preinstalled cabling is accessible through the central opening thereof) and leveled. The base plate **1202** is then fastened thereto, which in turn supports the back plate **1920**, front plate **1902**, doorframe **1940**, and beam **1942**. Service panel **1904** is removably secured to the front plate **1902** and features a locking mechanism **1906**. Back plate cap **1922** is disposed on top of the back plate **1920** and front plate **1902**, and is in turn secured to the bot plate **1918** which supports one or more devices. As shown, the devices comprise card reader **1912** and one or more plastic windows behind which further devices (such as an RFID reader and/or optical reader **1914**, **1916**) may be disposed. Upper back plate **1922** is placed above back plate **1920** while cover **1910** (which is generally similar to hood **110**) covers the bot plate **1918** and back cover **1920**.

Fifth Embodiment

FIGS. **20** through **26** illustrate a fifth embodiment of a terminal **2100** that is generally similar to terminal **100** except as otherwise noted herein.

As shown, components of the terminal **2100** are affixed together using one or more fasteners **2102**. The terminal **2100** comprises a single set of vents **126** located proximate the lower edge **130b** of the back plate **120**. These vents **126** draw in ambient air, which then is heated and rises through the terminal before being expelled through top vent **802**.

Sixth Embodiment

FIGS. **27** through **43** illustrate a sixth embodiment of a terminal **2800** that is generally similar to terminals **100**, **1900**, and **2100** except as otherwise noted herein. Unless otherwise noted, similarly labeled elements refer to similar components (i.e., front plate **102**, **2802**).

As depicted in FIGS. **27** through **43**, the terminal **2800** comprises a front plate **2802**, front access panel **2804**, hood **2810**, inner panel **2808**, back plate **2820**, and upper back plate **2822** that cooperatively forms the exterior surface of the terminal **2800**. As is clear to one of skill in the art, in alternative embodiments, the foregoing components may each comprise multiple elements or may be combined into single physical structures. For example, in an embodiment, the front plate **2802** comprises multiple plates that are separately formed before being jointed together and similarly the back plate **2820** and upper back plate **2822** are jointed together to form the back face of the terminal **2800**. Other alternative arrangements will be clear to one of skill in the art from the present disclosure.

One or more devices may be disposed along inner panel **2808**. In the embodiment depicted in FIGS. **27** and **28**, a scanner/reader **2816** for reading the barcode of a card, paper, or mobile device; an RFID reader **2814** for optically reading a radio-frequency ID (RFID) and/or near field communication (NFC); and a credit card reader and PIND pad **2812** are each disposed on the inner panel **2808**.

As depicted in FIGS. **27** through **43**, the inner panel **2808** further comprises a display (or screen) **2824**. In the preferred embodiment, the display **2824** is located proximate the top left corner of the inner panel **108**. In the preferred embodiment, the display **2824** is located above the first card reader **112**, second card reader **114**, and barcode reader **116**.

As depicted in FIG. **40**, the inner panel is comprised of a cutout **2818** with portions for placement of the first card

reader **2812**, second card reader **2814**, and barcode reader **2816**. These devices are held in place by a frame **4102**. Further, a display box **4112** is located within the housing of the hood **2810** and connects to the interior of the cutoff **2818**. The display box **4112** comprises a display screen **2824** as well as housing for the components necessary for the display screen **2824** to function.

As will be clear to one of skill in the art, and as shown in FIGS. **29** through **43**, other arrangements and/or combinations of devices may also or instead be disposed on the inner panel **2808**. Each of the devices **2812**, **2814**, **2816**, and **2824** are coupled to a controller **1002** and power supply (not shown), which may be provided through electrical means and/or a battery, such as through wiring (not shown, but as described in the main embodiment **706**). A central channel (not shown, but as described in the main embodiment **704**) in the terminal **2800** permits the routing of cabling **706** between components as well as the flow of air through the terminal **2800**. In one embodiment, one or more speakers (not shown, but as described in the main embodiment **702**) and/or microphones are disposed along the inner panel **2808**. In embodiments, channels **1602** are configured to receive corresponding protrusions on the hood **2810** so as to enable the hood **2810** to be removably connected to the body of the terminal **2800**.

Installation and Maintenance

The terminal **2800** is configured so as to permit ready access to the interior of the terminal **2800** and permit ready installation **2800** with limited or no use of specialized tools.

In the preferred embodiment, the terminal **2800** comprises an easily movable or removable access panel **2804**, whereby the access panel **2804** may be moved or removed in order to gain access to the interior of the terminal **2804**.

In the embodiment depicted in FIG. **40**, a bottom bracket **4104** connects to the lower edge of the front panel **2804** and interlocks with a corresponding channel **4106** on a lower interior edge of the front plate **2802**. A locking protrusion **4108** is movably connected to the keyhole **2806** such that the locking protrusion **4108** rotates as the keyhole **2806** is turned. The locking protrusion **4108** corresponds to a corresponding locking portion **4110**. With the access panel **2804** inserted in the front plate **2802** and the locking protrusion **4108** in an upwards extending position, the access panel **2804** is secured to the front plate **2802** and cannot be removed. To remove the access panel **2804**, the keyhole is actuated and rotated, causing the locking protrusion **4108** to rotate and disengage from the locking portion **4110**. The access panel **2804** can then be tilted away from the terminal **2800** and entirely removed, leaving an opening in the front panel **2802** through which the interior of the terminal **2800** may be accessed. This ensures the interior of the terminal **2800** cannot be accessed without actuating the keyhole **2806** (i.e., by using the corresponding key). By using a corresponding key, technicians can readily access the interior of the terminal **2800**.

In alternative embodiments, no hinges are used to movably connect the access panel **2804** to the front plate **2802**; the lower portion of the access panel **2804** is secured by a lock channel to allow for easy removal of the panel **2804** completely. Other structures for providing a movable or removable access panel **2804** will be clear to one of skill in the art in view of the instant disclosure and are also expressly contemplated hereby.

In an embodiment, the interior of the terminal **2800** further comprises a plurality of beams **4114** for additional attachments, such as the controller **1002**. The beams **4114**

allow for the controller **1002** or other attachments to be positioned in alternative locations.

In the preferred embodiment, the terminal **2800** further comprises a back access panel **2836** which is secured to the back plate **2820** by fasteners **2824**. The back access panel **2836** is removable to provide access to the interior of the terminal **2800** and/or to increase the circulation of ambient air through the terminal **2800**.

In one embodiment of the terminal **2800**, the covering of the terminal **2800** may be formed from multiple structural pieces that are bent, utilizing few welds and/or connectors. In such embodiment, the front plate **2802** and back plate **2820** are each formed from a single piece of stamped metal that is bent into a desired shape before and therefor fit together. Similarly, the hood **2810** is formed from a single piece of stamped metal before the inner baffle **1702** is welded to the hood.

Alternatively, the terminal **2800** may be comprised of several separate pieces stamped from multiple sheets of metal and secured with fasteners **2824**.

In the preferred embodiment, the terminal **2800** comprises a removable bottom plate **1202** (as described above) on which the entire terminal **2800** rests. Such configuration greatly improves the ease of installation of a terminal **2800**. In use, the bottom plate **1202** is positioned at a desired location and leveled. Inner opening **1206** may be located so as to allow cabling **702** to enter the terminal **2800**, including but not limited to pre-installed cabling **702** to provide mains power and/or communication connections to the terminal **2800**. Outer openings **1204** are configured to permit fasteners **2824** (such as screws and bolts) to be driven through the bottom of the plate **1202** to secure the bottom plate **1202** in place. In embodiments, openings **1204** are curved so as to allow the bottom plate **1202** to be turned (e.g., to avoid rebar or other embedded obstacles within the substructure) and are extended so as to allow fasteners to be placed at preferred locations. The raised edges **1208** of the plate **1202** and notches **1210** serve to cooperatively receive the bottom edge **2830b** of the back plate **2820** and/or front plate **2802** so as to allow the terminal **2800** to be secured in place. In embodiments, fasteners **2824** may be used to further secure the terminal **2800** to the bottom plate **1202**.

The removable bottom plate **1202** beneficially enables the bottom plate **1202** to be easily located and levelled without the need to lift, move, and/or adjust the larger terminal **2800** as a whole. Further, the ability to connect the terminal **2800** to the bottom plate **1202** without tools enables faster installation.

Passive Cooling

In a preferred embodiment, the terminal **2800** is passively cooled. Such configuration reduces the number of components (and thus component cost, installation complexity, and points of failure for the terminal) as well as the operating cost of the terminal **2800** (e.g., by reducing the ongoing power requirements by eliminating the need to expend energy cooling the terminal **2800**). Embodiments contemplated hereby incorporate one or more of the following features.

In the embodiments depicted in FIGS. 27 through 43, the top surface **2828** of the hood **2810** is angled. As will be clear to one of skill in the art, the top surface **2828** is the portion of the terminal that receives the greatest amount of surface radiation. Angling the top surface **128** serves both to aid in expelling water from the top surface **2828** prior to its evaporation and in minimizing the thermal effect of solar radiation striking the top surface **2828**. The angle of the "tilt" (α) of the top surface **2828** may be determined based on the

intended geographic location of the terminal **2800**. In embodiments, the tilt is greater than 26 degrees from the horizontal. In alternative embodiments, the tilt is between approximately 40 and 50 degrees from the horizontal.

In contrast, the front plate **2802** and back plate **2822** are substantially vertical, thereby minimizing the area receiving solar radiation on the surfaces thereof.

As depicted in FIGS. 27 through 43, the hood **2810** extends over the inner panel **2808** such that the hood **2810** provides shading to the inner panel **2808**. In embodiments, the hood **2810** is adjustable to increase the amount of shading provided to the inner panel **2808**.

To further aid in dissipating solar radiation, the exterior surface of the terminal **2800** may be a light or reflective color, such that solar radiation is reflected away from the terminal **2800**. In the preferred embodiment, the hood **2810** and front panel **2802** are painted or otherwise colored to the RAL-9010 shade of white while the inner panel **2808** is painted or otherwise colored to the RAL-9006 shade of aluminum grey.

In addition, vents **2826** are located along the back plate **2820**. In the preferred embodiment, a set of vents **2826** are disposed proximate the lower edge **2830b** of the back plate **2820**. In the embodiment depicted, the vents **2826** serve as air intake ports, allowing ambient air to enter the terminal **2800** therethrough. As the ambient air is warmed, it rises through the interior of the terminal **2800** before being expelled through a top vent **802** disposed between the top edge **2832** of the interior panel **2808** and the hood **2810**. Vents **2826**, **802** are comprised of openings to permit the passage of air.

The hood **2810** is angled such that the front edge **2834a** is wider than the back edge **2834b**. In the preferred embodiment, the angle between the sides of the hood **2810** (σ) is between approximately 40 and 50 degrees. This aids in creating a chimney effect whereby air exiting through the top vent **802** creates a lower pressure zone that accelerates the flow of air through the terminal **2800** (and along the surfaces of the electrical components disposed therein), increasing the effective cooling rate. In embodiments, the top vent **802** extends along the entire edge **2832** of the interior panel **2808**. In alternative embodiments, one or more vents **802** are located along the top edge **2832**.

Water Redistribution

In embodiments, the exterior of the terminal **2800** is made from a waterproof material. In further embodiments, the terminal **2800** is specifically designed to ensure water entering the terminal **2800** is expelled without accessing electrical or other sensitive components. In an embodiment, a back channel **708** is angled across the interior surface of the inner panel **2808** and protrudes horizontally towards the back of the terminal **2800**. The back channel **708** functions as a gutter, catching water that gains entrance to the terminal **2800** around the edges of the inner panel **108** to drain out of the terminal **2800**. In embodiments, water may enter through a top opening **802** and is expelled through a bottom opening **804** disposed between the inner panel **2808** and hood **2810**.

The foregoing examples have been provided merely for the purpose of explanation and are in no way to be construed as limiting of the present invention disclosed herein. While the invention has been described with reference to various embodiments, it is understood that the words, which have been used herein, are words of description and illustration, rather than words of limitation. Further, although the invention has been described herein with reference to particular means, materials, and embodiments, the invention is not intended to be limited to the particulars disclosed herein;

11

rather, the invention extends to all functionally equivalent structures, methods, and uses, such as are within the scope of the appended claims. Those skilled in the art, having the benefit of the teachings of this specification, may affect numerous modifications thereto and changes may be made without departing from the scope and spirit of the invention in its aspects.

Any other undisclosed or incidental details of the construction or composition of the various elements of the disclosed embodiment of the present invention are not believed to be critical to the achievement of the advantages of the present invention, so long as the elements possess the attributes needed for them to perform as disclosed. Certainly, one skilled in the art would be able to conceive of a wide variety of alternatives and system configurations and successful combinations thereof. The selection of these and other details of construction are believed to be well within the ability of one of even rudimentary skills in this area, in view of the present disclosure. Illustrative embodiments of the present invention have been described in considerable detail for the purpose of disclosing a practical, operative structure whereby the invention may be practiced advantageously. The designs described herein are intended to be exemplary only. The novel characteristics of the disclosure may be incorporated in other structural forms without departing from the spirit and scope of the invention. The invention encompasses embodiments both comprising and consisting of the elements described with reference to the illustrative embodiments. Unless otherwise indicated, all ordinary words and terms used herein shall take on their customary meaning. All technical terms shall take on their customary meaning as established by the appropriate technical discipline utilized by those normally skilled in that particular art area.

The invention claimed is:

1. A parking terminal comprising:

- a bottom plate;
- a body;
- a first vent located on a back plate of the body of the terminal;
- a front panel;
- a hood; and
- an electronic access control mechanism that, based on interaction with the parking terminal by a human, transmits electronic instructions to one or more hardware components that regulate vehicle traffic with respect to a parking facility, the one or more hardware components not physically connected to the parking terminal;

wherein the bottom plate is configured to be secured to a surface, the body is removably attached to the bottom plate and the front panel and the hood are supported by the body.

2. The parking terminal of claim 1, wherein the hood comprises a top surface which forms an angle α from a horizontal axis.

3. The parking terminal of claim 2, wherein the angle α is approximately equal to or at least 26 degrees.

4. The parking terminal of claim 2, wherein the angle α is between approximately 40 and approximately 50 degrees.

- 5. The parking terminal of claim 2, further comprising:
 - a drain outlet located proximate a first side of the front panel; and
 - a channel disposed on an interior surface of the front panel.

12

6. The parking terminal of claim 5, further comprising a baffle located on an inner surface of the hood proximate the front panel.

7. The parking terminal of claim 1, the terminal further comprising a second vent located between the hood and the front panel.

8. The parking terminal of claim 1, the terminal further comprising a frame, wherein the one or more hardware components are removably attached to the frame and wherein the frame is supported by the body.

9. The parking terminal of claim 8, wherein the one or more devices disposed on the front panel comprise a bar code reader, an RFID reader, and a credit card reader.

10. The parking terminal of claim 9, wherein the front panel further comprises a display.

11. The parking terminal of claim 10, wherein the display comprises a touch screen.

12. The parking terminal of claim 1, wherein the hood comprises a first side, a second side, and a top surface and the first side and the second side form an angle σ therebetween and wherein the angle σ is between approximately 40 and approximately 50 degrees.

13. The parking terminal of claim 1, further comprising:

- an opening in the bottom plate; and
- a passage in the body aligned with the opening.

14. A method of installing a parking terminal comprising a bottom plate, a body, a first vent located on a back plate of the body of the terminal a front plate adapted to receive one or more devices, and a hood, the method comprising steps of:

- selecting a location for installation of the terminal;
- leveling a surface at the location;
- positioning the bottom plate at the location;
- securing the bottom plate in place at the location;
- aligning a lower edge of the body with the bottom plate;
- securing the body to the bottom plate;
- securing the front plate to the body; and
- securing the hood to the body, wherein the location is a parking facility, and wherein the parking terminal comprises an electronic access control mechanism that, based on interaction with the parking terminal by a human, transmits electronic instructions to one or more hardware components that regulate vehicle traffic with respect to a parking facility, the one or more hardware components not physically connected to the parking terminal.

15. The method of claim 14, wherein the bottom plate comprises a vertical support extending around at least a portion of a perimeter of the bottom plate and the step of aligning the lower edge of the body with the bottom plate comprises aligning the body with the vertical support.

16. The method of claim 15, wherein the bottom plate comprises an opening and the body comprises a passageway, the method further comprising the steps of prior to positioning the bottom plate at the location, installing cabling at the location and routing the cabling through the opening in the bottom plate;

- prior to aligning the lower edge of the body with the bottom plate, routing the cabling through the passageway within the terminal body;
- attaching the cabling to a controller within the body; and
- communicatively coupling the controller to the one or more devices.

17. A method of servicing a parking terminal comprising a bottom plate, a body having a front and rear access panel, a first vent located on a back plate of the body of the

13

terminal, a front panel adapted to receive one or more devices, and a hood, the method comprising steps of:

opening the front access panel by rotating a key in a key slot located in the front access panel;

accessing an interior of the body;

5

closing the front access panel; and

locking the front access panel, wherein the location is a

parking facility, and wherein the parking terminal com-

prises an electronic access control mechanism that,

based on interaction with the parking terminal by a

10

human, transmits electronic instructions to one or more

hardware components that regulate vehicle traffic with

respect to a parking facility, the one or more hardware

components not physically connected to the parking

terminal.

15

18. The method of claim 17, further comprising the steps of:

removing the hood from the body;

removing at least one of the one or more devices; and

replacing the hood on the body.

20

19. The method of claim 17, further comprising the steps of detaching the body from the bottom plate;

accessing cabling located proximate the bottom plate; and

reattaching the body on the bottom plate.

* * * * *

25

14